

UNIT-1

Introduction to Internet of Things

Internet of things (IoT)

The Internet of things (IoT) is the inter-networking of physical devices, vehicles (also referred to as “connected devices” and “smart devices”), buildings, and other items embedded with electronics, software, sensors, actuators, and network connectivity which enable these objects to collect and exchange data.

Characteristics:

Things-related services: The IoT is capable of providing thing-related services within the constraints of things, such as privacy protection and semantic consistency between physical things and their associated virtual things

Connectivity: Things in I.O.T. should be connected to the infrastructure, without connection nothing makes sense.

Intelligence: Extraction of knowledge from the generated data is important, sensor generate data and this data and this data should be interpreted properly.

Scalability: The no. of things getting connected to the I.O.T. infrastructure is increased day by day. Hence, an IOT setup shall be able to handle the massive expansion.

Unique Identity: Each IOT device has an I.P. address. This identity is helpful in tracking the equipment and at times to query its status.

Dynamic and Self-Adapting: The IOT device must dynamically adopt itself to the changing context. Assume a camera meant for surveillance, it may have to work in different conditions and at different light situations (morning, afternoon, night).

Heterogeneity: The devices in the IoT are heterogeneous as based on different hardware platforms and networks. They can interact with other devices different networks.

Safety: Having got all the things connected with the Internet possess a major threat, as our personal data is also there and it can be tampered with, if proper safety measures are not taken.

Application areas of IoT:

Smart Home: The smart home is one of the most popular applications of IoT. The cost of owning a house is the biggest expense in a homeowner's life. Smart homes are promised to save the time, money and energy.

Smart cities: The smart city is another powerful application of IoT. It includes smart surveillance, environment monitoring, automated transformation, urban security, smart traffic management, water distribution, smart healthcare etc.

Wearables: Wearables are devices that have sensors and software installed which can collect data about the user which can be later used to get the insights about the user. They must be energy efficient and small sized.

Connected cars: A connected car is able to optimize its own operation, maintenance as well as passenger's comfort using sensors and internet connectivity.

Smart retail: Retailers can enhance the in-store experience of the customers using IoT. The shopkeeper can also know which items are frequently bought together using IoT devices.

Smart healthcare: People can wear the IoT devices which will collect data about user's health. This will help users to analyze themselves and follow tailor-made techniques to combat illness. The doctor also doesn't have to visit the patients in order to treat them.

IoT Categories

IOT can be classified into two categories:

1. Consumer IoT(CIOT): The Consumer IoT refers to the billions of physical personal devices, such as smartphones, wearables, fashion items and the growing number of smart home appliances, that are now connected to the internet, collecting and sharing data.

A Consumer IoT network typically entails few consumer devices, each of which has a limited lifetime of several years.

The common connectivity used in this kind of solutions are Bluetooth, WiFi, and ZigBee. These technologies offer short-range communication, suitable for applications deployed in limited spaces such as houses, or small offices.

2. industrial internet of things (IIoT): It refers to interconnected sensors, instruments, and other devices networked together with computers' industrial applications, including manufacturing and energy management. This connectivity allows for data collection, exchange, and analysis, potentially facilitating improvements in productivity and efficiency as well as other economic ben

BASELINE TECHNOLOGIES

There are various baseline technologies that are very closely related to IOT, They include: Machine-to-Machine (M2M), Cyber-Physical Systems (CPS), Web Of Things(WOT)

a) Machine-to-Machine (M2M):

- Machine-to-Machine (M2M) refers to networking of machines (or devices) for the purpose of remote monitoring and control and data exchange.
- An M2M area network comprises of machines (or M2M nodes) which have embedded network modules for sensing, actuation and communicating various communication protocols can be used for M2M LAN such as ZigBee, Bluetooth, M-bus, Wireless M-Bus etc., These protocols provide connectivity between M2M nodes within an M2M area network.
- The communication network provides connectivity to remote M2M area networks. The communication network provides connectivity to remote M2M area network.
- The communication network can use either wired or wireless network (IP based). While the M2M are networks use either proprietary or non-IP based communication protocols, the communication network uses IP-based network. Since non-IP based protocols are used within M2M area network, the M2M nodes within one network cannot communicate with nodes in an external network.
- To enable the communication between remote M2M are network, M2M gateways are used

b) Cyber-Physical systems:

Cyber-Physical Systems (CPS) are integrations of computation, networking, and physical processes. Embedded computers and networks monitor and control the physical processes, with feedback loops where physical processes affect computations and vice versa.

In cyber-physical systems, physical and software components are deeply intertwined, able to operate on different spatial and temporal scales, exhibit multiple and distinct behavioural modalities, and interact with each other in ways that change with context.

c) **Web of Things:** web of things is a term used to describe approaches, software architectural style of programming patterns that allow real world objects to be part of WWW. The major portion of the WoT specification is the Thing Description. Thing is an abstract representation of a physical or virtual entity. A Thing Description includes the metadata and interfaces of a Thing in a standardized way, with the aim to make the Thing able to communicate with other Things in a heterogeneous world.

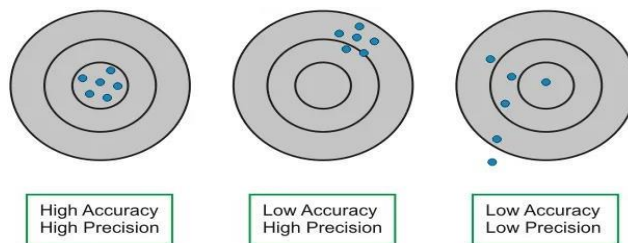
SENSOR

Sensor is a device used for the conversion of physical events or characteristics into the electrical signals. This is a hardware device that takes the input from environment and gives to the system by converting it.

For example, a thermometer takes the temperature as physical characteristic and then converts it into electrical signals for the system.

Characteristics of Sensors

1. **Range:** It is the minimum and maximum value of physical variable that the sensor can sense or measure. For example, a Resistance Temperature Detector (RTD) for the measurement of temperature has a range of -200 to 800°C .
2. **Span:** It is the difference between the maximum and minimum values of input. In above example, the span of RTD is $800 - (-200) = 1000^{\circ}\text{C}$.
3. **Accuracy:** The error in measurement is specified in terms of accuracy. It is defined as the difference between measured value and true value. It is defined in terms of % of full scale or % of reading.
4. **Precision:** It is defined as the closeness among a set of values. It is different from accuracy.



5. **Linearity:** Linearity is the maximum deviation between the measured values of a sensor from ideal curve.

6. **Hysteresis:** It is the difference in output when input is varied in two ways- increasing and decreasing.

7. **Resolution:** It is the minimum change in input that can be sensed by the sensor.

8. **Reproducibility:** It is defined as the ability of sensor to produce the same output when same input is applied.

9. **Repeatability:** It is defined as the ability of sensor to produce the same output every time when the same input is applied and all the physical and measurement conditions kept the same including the operator, instrument, ambient conditions etc.

10. **Response Time:** It is generally expressed as the time at which the output reaches a certain percentage (for instance, 95%) of its final value, in response to a step change of the input.

Classification of sensors:

- Sensors based on the power requirement sensor is classified into two types: Active Sensors, Passive Sensors.

Active Sensors: Does not need any external energy source but directly generates an electric signal in response to the external.

Example: Thermocouple, Photodiode, Piezoelectric sensor.

Passive Sensors: The sensors require external power called excitation signal. Sensors modify the excitation signal to provide output.

Example: Strain gauge.

➡ Sensors based on output sensor is classified into two types: Analog Sensors, Digital Sensors.

Analog Sensors

- Analog Sensors produces a continuous output signal or voltage which is generally proportional to the quantity being measured.
- Physical quantities such as Temperature, speed, Pressure, Displacement, Strain etc. are all analog quantities as they tend to be continuous in nature.
- For example, the temperature of a liquid can be measured using a thermometer or thermocouple (e.g. in geysers) which continuously responds to temperature changes as the liquid is heated up or cooled down.

Digital Sensors

- Digital Sensors produce discrete output voltages that are a digital representation of the quantity being measured.
- Digital sensors produce a binary output signal in the form of a logic "1" or a logic "0" , ("ON" or "OFF").
- Digital signal only produces discrete (non-continuous) values, which may be output as a signal "bit" (serial transmission), or by combining the bits to produce a signal "byte" output (parallel transmission).

➡ Based on type of data measured sensor is classified into two types: Scalar Sensors and Vector Sensors.

Scalar Sensors

- Scalar Sensors produce output signal or voltage which generally proportional to the magnitude of the quantity being measured.
- Physical quantities such as temperature, color, pressure, strain, etc. are all scalar quantities as only their magnitude is sufficient to convey an information.
- For example the temperature of a room can be measured using thermometer or thermocouple, which responds to temperature changes irrespective of the orientation of the sensor or its direction.

Vector Sensors

- Vector Sensors produce output signal or voltage which generally proportional to the magnitude, direction, as well as the orientation of the quantity being measured.
- Physical quantities such as sound, image, velocity, acceleration, orientation, etc. are all vector quantities, as only their magnitude is not sufficient to convey the complete information.
- For example, the acceleration of a body can be measured using an accelerometer, which gives the components of acceleration of the body with respect to the x,y,z coordinate axes.

ACTUATOR

Actuator is a device that converts the electrical signals into the physical events or characteristics. It takes the input from the system and gives output to the environment. For example, motors and heaters are some of the commonly used actuators.

Types of Actuators

1. Hydraulic Actuators: Hydraulic actuators operate by the use of a fluid-filled cylinder with a piston suspended at the centre. Commonly, hydraulic actuators produce linear movements, and a spring is attached to one end as a part of the return motion. These actuators are widely seen in exercise equipment such as steppers or car transport carriers.

2. Pneumatic Actuators: Pneumatic actuators are one of the most reliable options for machine motion. They use pressurized gases to create mechanical movement. Many companies prefer pneumatic-powered actuators because they can make very precise motions, especially when starting and stopping a machine. Examples of equipment that uses pneumatic actuators include: Bus brakes, Exercise machines, Vane motors, Pressure sensors

3. Electric Actuators : Electrical actuators, as you may have guessed, require electricity to work. Well-known examples include electric cars, manufacturing machinery, and robotics equipment. Similar to pneumatic actuators, they also create precise motion as the flow of electrical power is constant.

4. Thermal and Magnetic Actuators : Thermal and magnetic actuators usually consist of shape memory alloys that can be heated to produce movement. The motion of thermal or magnetic actuators often comes from the Joule effect, but it can also occur when a coil is placed in a static magnetic field. The magnetic field causes constant motion called the Laplace-Lorentz force. Most thermal and magnetic actuators can produce a wide and powerful range of motion while remaining lightweight.

5. Mechanical Actuators : Some actuators are mostly mechanical, such as pulleys or rack and pinion systems. Another mechanical force is applied, such as pulling or pushing, and the actuator will leverage that single movement to produce the desired results. For instance, turning a single gear on a set of rack and pinions can mobilize an object from point A to point B. The tugging movement applied on the pulley can bring the other side upwards or towards the desired location.

6. Soft Actuators: Soft actuators (e.g. polymer based) are designed to handle fragile objects like fruit harvesting in agriculture or manipulating the internal organs in biomedicine.

They typically address challenging tasks in robotics. Soft actuators produce flexible motion due to the integration of microscopic changes at the molecular level into a macroscopic deformation of the actuator materials.

IOT COMPONENTS

Four fundamental components of IoT system, which tells us how IoT works.

i. Sensors/Devices

First, sensors or devices help in collecting very minute data from the surrounding environment. All of this collected data can have various degrees of complexities ranging from a simple temperature monitoring sensor or a complex full video feed.

A device can have multiple sensors that can bundle together to do more than just sense things. For example, our phone is a device that has multiple sensors such as GPS, accelerometer, camera but our phone does not simply sense things.

ii. Connectivity

Next, that collected data is sent to a cloud infrastructure but it needs a medium for transport.

The sensors can be connected to the cloud through various mediums of communication and transports such as cellular networks, satellite networks, Wi-Fi, Bluetooth, wide-area networks (WAN), low power wide area network and many more.

iii. Data Processing

Once the data is collected and it gets to the cloud, the software performs processing on the acquired data.

This can range from something very simple, such as checking that the temperature reading on devices such as AC or heaters is within an acceptable range. It can sometimes also be very complex, such as identifying objects (such as intruders in your house) using computer vision on video.

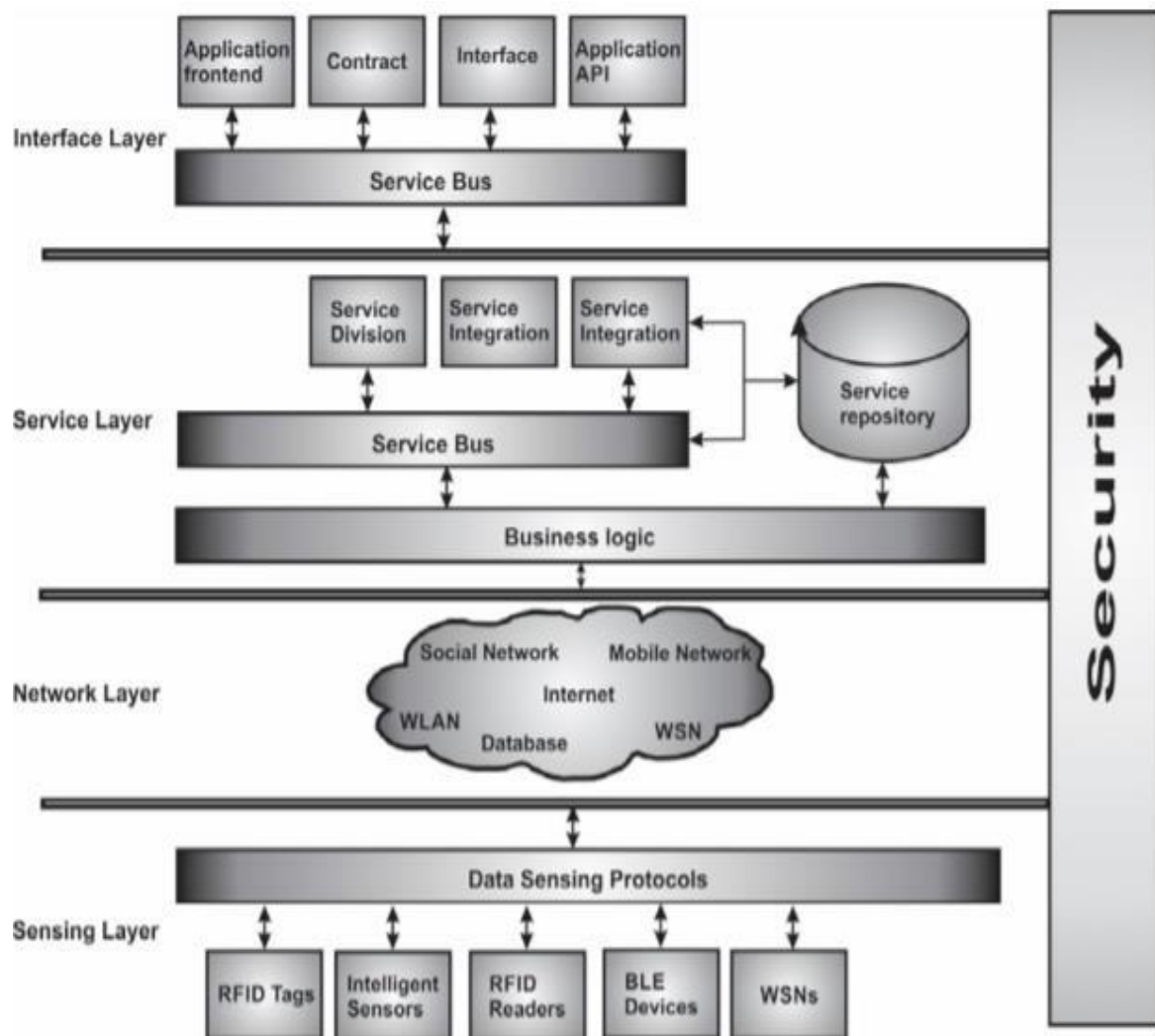
iv. User Interface

Next, the information made available to the end-user in some way. This can achieve by triggering alarms on their phones or notifying through texts or emails.

Also, a user sometimes might also have an interface through which they can actively check in on their IOT system. For example, a user has a camera installed in his house, he might want to check the video recordings and all the feeds through a web server.

Service Oriented Architecture of IoT

SOA can also use to support IoT as a main contributing technology in devices or heterogeneous systems.



Service-Oriented Architecture for IoT technologies.

1. Sensing Layer: IoT can be defined as a worldwide interconnected network, where things or devices are controlled remotely. Interconnected things or devices are becoming easier, as more and more things are furnished with sensors and RFID technologies.

2. Networking Layer: Networking Layer is responsible to connect all devices or things together so that they can be able to share the information with each other over the Internet. Moreover, network layer also collects data and information from the present IT infrastructure for example ICT systems, power grids, business systems, healthcare systems, and transportation systems.

3. Service Layer: This layer depends upon the technology used on the middleware layer which is responsible for functionalities incorporated between applications and services in IoT. This middleware technology also provides a cost-effective and efficient platform for IoT and this platform including software and hardware components which can be reused when needed.

4. Interface Layer: The core responsibility of the interface layer has also simplified the interconnection and management of things. Interface specific profile can be defined as the subset of services that support interaction with the application used in a network

Challenges for IoT

1. Security: Security is the most significant challenge for the IoT. Increasing the number of connected devices increases the opportunity to exploit security vulnerabilities, as do poorly designed devices, which can expose user data to theft by leaving data streams inadequately protected and in some cases people's health and safety can be put at risk.

2. Privacy: The IoT creates unique challenges to privacy, many that go beyond the data privacy issues that currently exist. Much of this stems from integrating devices into our environments without us consciously using them. This is becoming more prevalent in consumer devices, such as tracking devices for phones and cars as well as smart televisions.

3. Scalability: Billions of internet-enabled devices get connected in a huge network, large volumes of data are needed to be processed. The system that stores, analyses the data from these IoT devices needs to be scalable.

4. Interoperability: Technological standards in most areas are still fragmented. These technologies need to be converged. Which would help us in establishing a common framework and the standard for the IoT devices. As the standardization process is still lacking, interoperability of IoT with legacy devices should be considered critical. This lack of interoperability is preventing us to move towards the vision of truly connected everyday interoperable smart objects.

5. Bandwidth: Connectivity is a bigger challenge to the IoT than you might expect. As the size of the IoT market grows exponentially, some experts are concerned that bandwidth-intensive IoT applications such as video streaming will soon struggle for space on the IoT's current server-client model.

6. Standards: Lack of standards and documented best practices have a greater impact than just limiting the potential of IoT devices. Without standards to guide manufacturers, developers sometimes design products that operate in disruptive ways on the Internet without much regard to their impact. If poorly designed and configured, such devices can have negative consequences for the networking resources they connect to and the broader Internet.

7. Regulation: The lack of strong IoT regulations is a big part of why the IoT remains a severe security risk, and the problem is likely to get worse as the potential attack surface expands to include ever more crucial devices. When medical devices, cars and children's toys are all connected to the Internet, it's not hard to imagine many potential disaster scenarios unfolding in the absence of sufficient regulation.

What is the Internet of Things?



In the IoT, things are expected to become active participants in business, information and social processes

In IoT, things are enabled to interact and communicate among themselves and with the environment by exchanging data and information “sensed” about the environment

Things should react autonomously to the “real/physical world” events with or without direct human intervention



IoT + Machine Learning

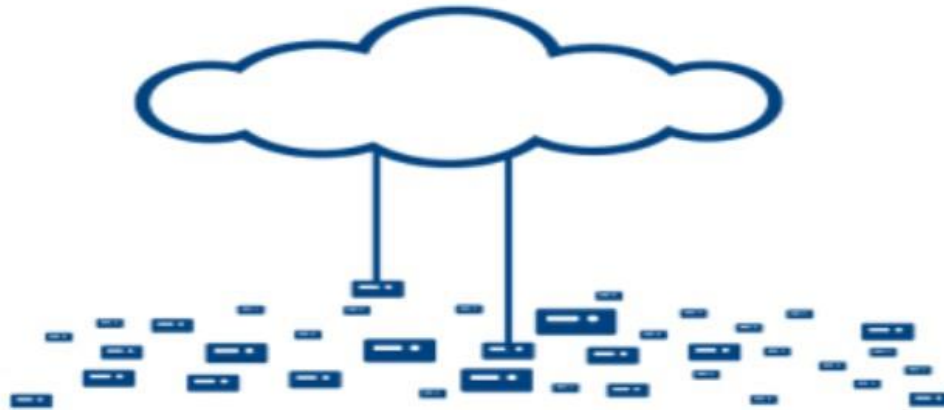
+ AI



Changing
the
Paradigm
of
Technology

AI and IoT: Together for a Better Future

- Artificial intelligence and the Internet of Things are among the hottest topics in recent times, gathering a lot of attention from technology enthusiasts.
- IoT collects huge amounts of data from innumerable environments.
- AI uses data science to convert the gathered data into important information and uses analytics to the data for decision making.



IoT + Machine Learning:

Artificial Intelligence and the Internet of Things (IoT) is growing at a fast pace. We can see how much IoT and AI help us in everyday life as well as businesses.

Today companies are using AI, powered by machine learning to understand data and provide informative decisions better along with leveraging personalization. Virtual assistants like Siri, Google Assistant,

Cloud based voice assistants

- Alexa is the voice assistant from Amazon, which is used in products like Amazon Echo, Amazon Tap etc. There is a specific set of skills known as the Alexa Skills Kit (ASK) that can be modified and updated to personalise or improve certain skills.
- Siri from Apple Inc. is used in Apple Homepod which serves a similar purpose. † Google Assistant used in
- Google Home has additional features where it can recognise up to six different users and pull their respective details to converse with them.

- Smart Lights by Deako can be controlled remotely through smartphones and Alexa or Google Assistant. They are connected via the Internet and can receive software upgrades from time to time.
- Automotive AI by Affectiva is an in-cabin sensing AI that can be used in robo taxis and highly automated vehicles. It detects emotional and cognitive state of the occupants in the vehicle from their face and voice through in-cabin cameras and microphones.
- Skybell: Homeowners can enjoy a full, two-way conversation with visitors from their iPhone or iPad. The SkyBell Video Doorbell DBCAM alerts homeowners when visitors are in front of their door, whether or not the button is pressed. Now, homeowners can feel safe knowing they can view visitors at night in color HD.

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IoT + Machine Learning:

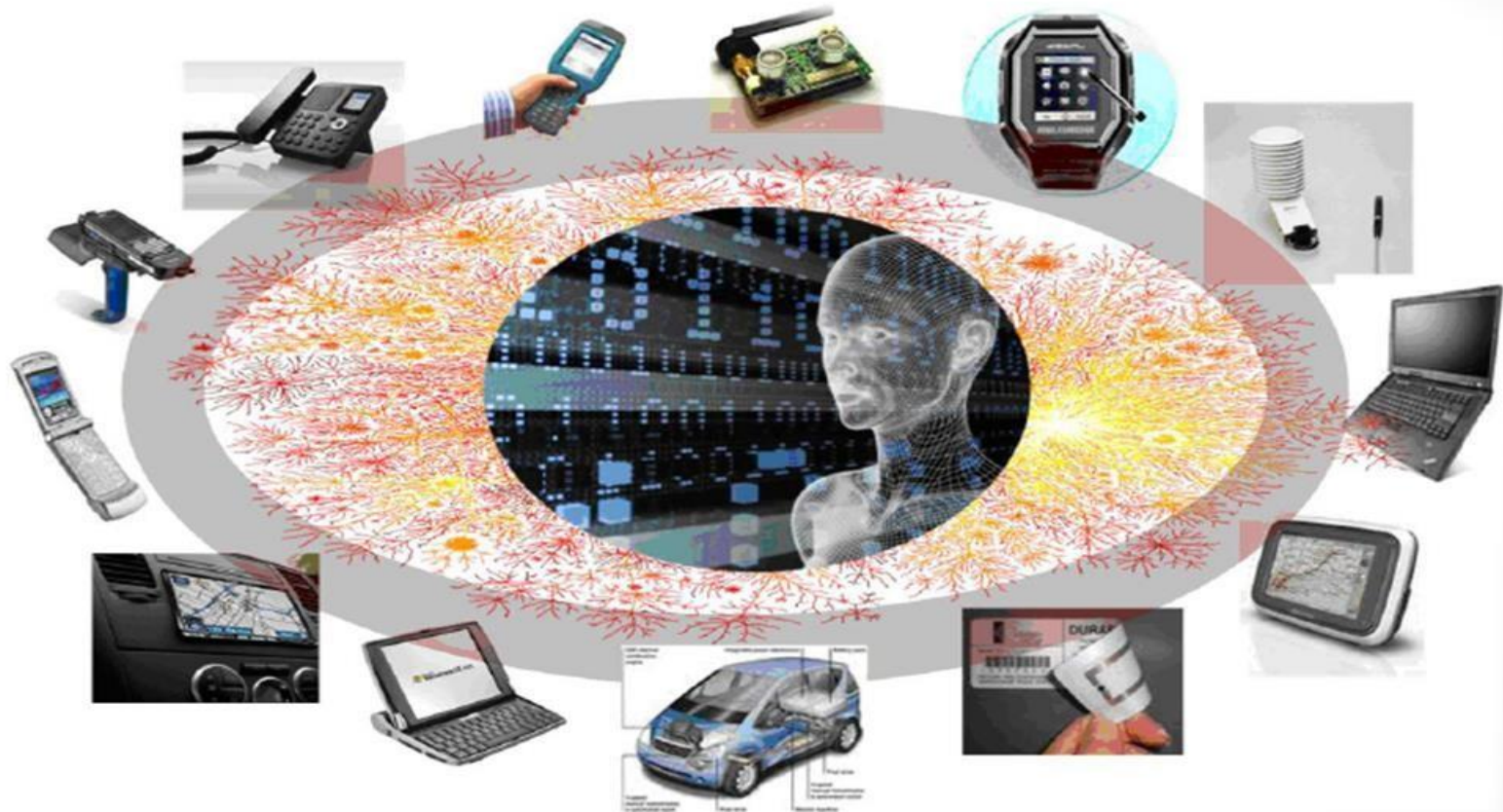
- According to research by Statista, connected IoT devices are estimated to be 30.9 billion units by 2025. This shows the amount of demand and applications IoT has in the market.
- **What is Machine Learning in AI ?**
- Machine learning is one of the critical components of AI, where the algorithms are developed to improve the performance of the software. It uses data collection to understand and provides users with informative decisions and personalized experiences.

IoT + Machine Learning:

Machine Learning and Artificial Intelligence in Retail Commerce:

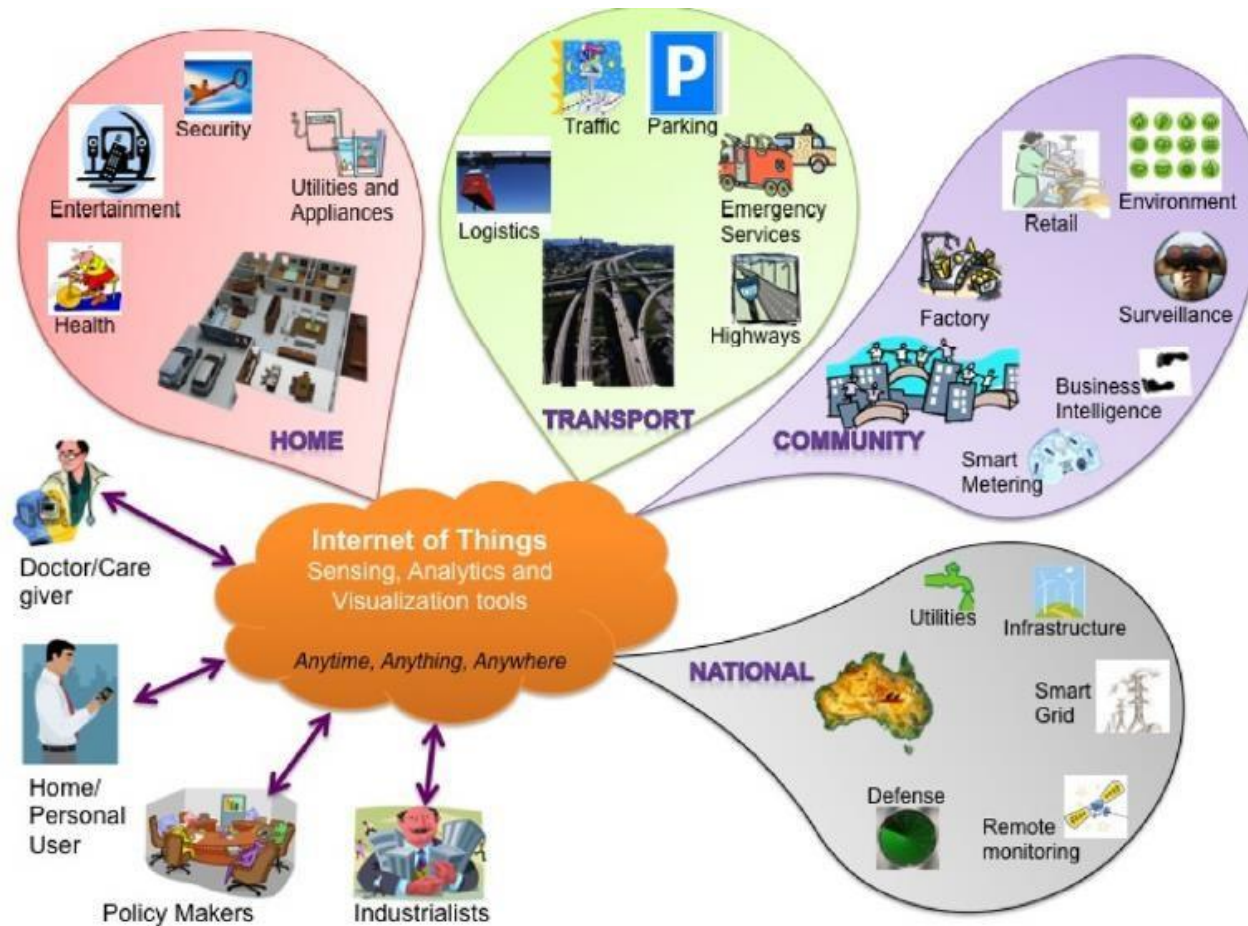
- eCommerce companies like Amazon, Wish, Myntra, etc., are using machine learning to understand their users better and provide personalization. With the help of big data, these companies can study their customers' purchase history, market trends, customers' spending patterns, habits, and preferences. A company studies this data, aggregates it, and uses it to make critical well-informed decisions.

Things Connected: communicated between physical world and information world



IOT end users & Applications

Kevin Ashton Coined IOT

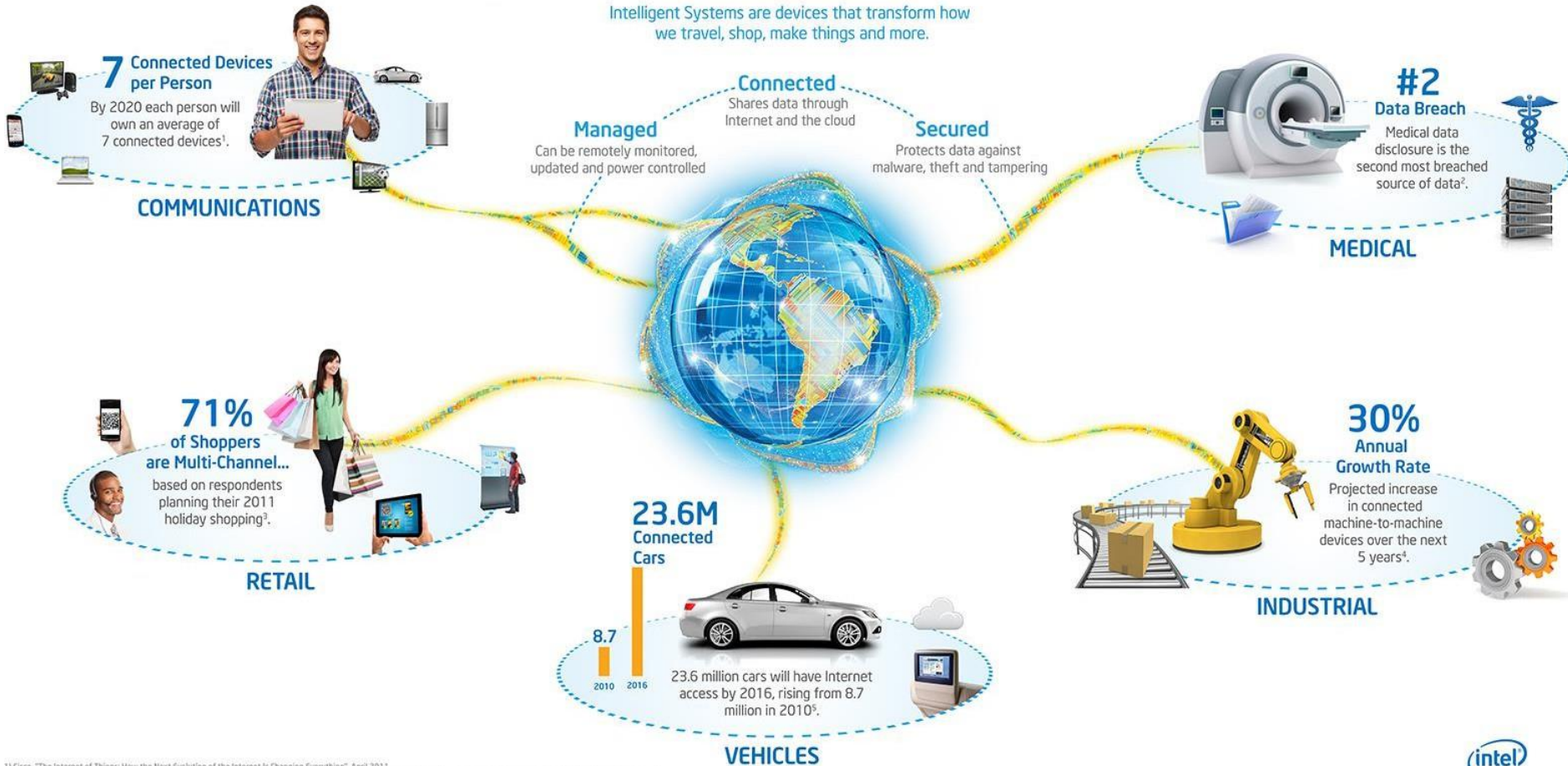


Opportunities

Intelligent Systems for a More Connected World

WHAT ARE INTELLIGENT SYSTEMS?

Intelligent Systems are devices that transform how we travel, shop, make things and more.



¹ Cisco, "The Internet of Things: How the Next Evolution of the Internet is Changing Everything", April 2011
² Bloor Research, "Security challenges in the US healthcare sector" white Paper, December 2010, <http://www.mcafee.com/us/resources/white-papers/wp-bloor-healthcare-security.pdf>
³ Deloitte U.S., 2011 Annual Holiday Survey, http://www.deloitte.com/assets/Content/Documents/Consumer/2011AnnualHolidaySurvey_2011_pr_102611.pdf
⁴ McKinsey Global Institute analysis, "Big data: The next frontier for innovation, competition, and productivity", June 2011
⁵ Wall Street Journal, <http://online.wsj.com/article/SB10001424052702304066504576349763614933844.html>, estimate from research firm, Frost & Sullivan

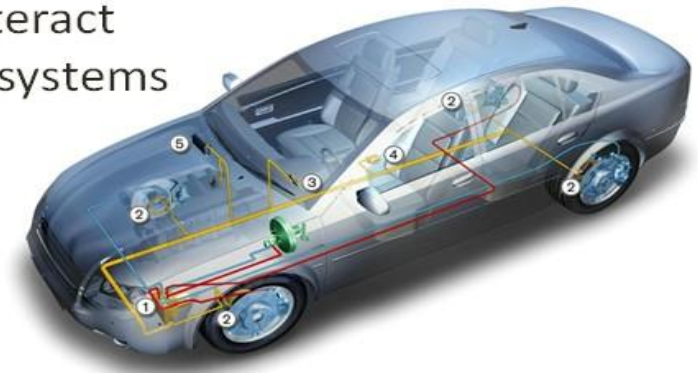
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History of Internet of things (IOT)

- **Kevin Ashton** (born 1968) is a British technology pioneer who cofounded the [Auto-ID Center](#) at the [Massachusetts Institute of Technology](#) (MIT), which created a global standard system for [RFID](#) and other sensors.^[1] He is known for coining the term "the [Internet of Things](#)"^[2] to describe a system where the Internet is connected to the physical world via ubiquitous sensors.

AI + IOT in transportation

- A network of sensors set up throughout a vehicle can interact with its surroundings to provide valuable feedback on local roads, weather and traffic conditions to the car driver, enabling adaptive drive systems to respond accordingly
- This may involve automatic activation of braking systems or speed control via fuel management systems. Condition and event detection sensors can activate systems to maintain driver and passenger comfort and safety through the use of airbags and seatbelt pre-tensioning
- Sensors for fatigue and mood monitoring based on driving conditions, driver behaviour and facial indicators can interact to ensure safe driving by activating warning systems or directly controlling the vehicle

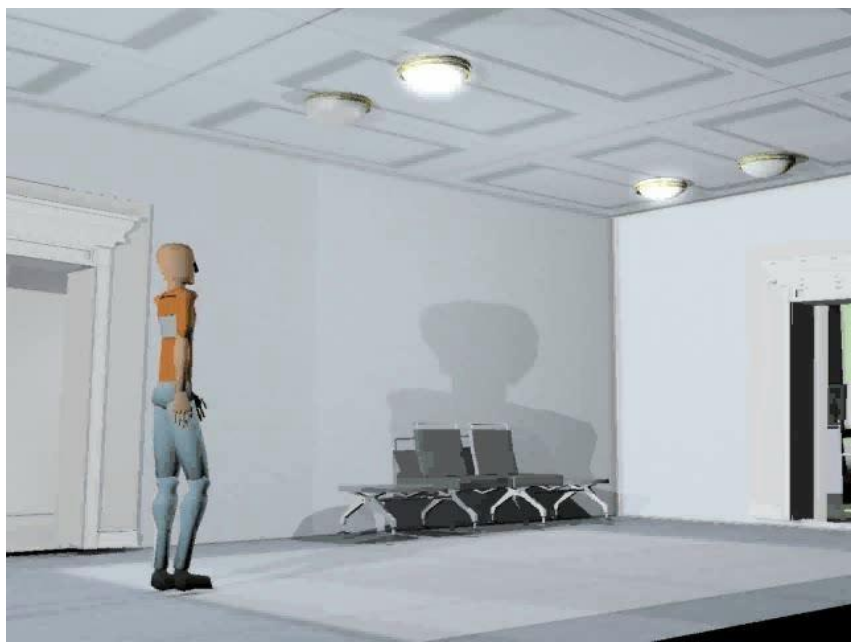


What is AIoT (artificial intelligence of things)?

- Artificial intelligence of things (AIoT) is the combination of artificial intelligence (AI) technologies and the internet of things (IoT) infrastructure. AIoT's goal is to create more efficient IoT operations, improve human-machine interactions and enhance [data management](#) and analytics.
- AIoT data can also be [processed at the edge](#), meaning the data from IoT devices is processed as close to these devices as possible to minimize the bandwidth needed to move data while avoiding possible delays to data analysis.

Applications OF AIOT

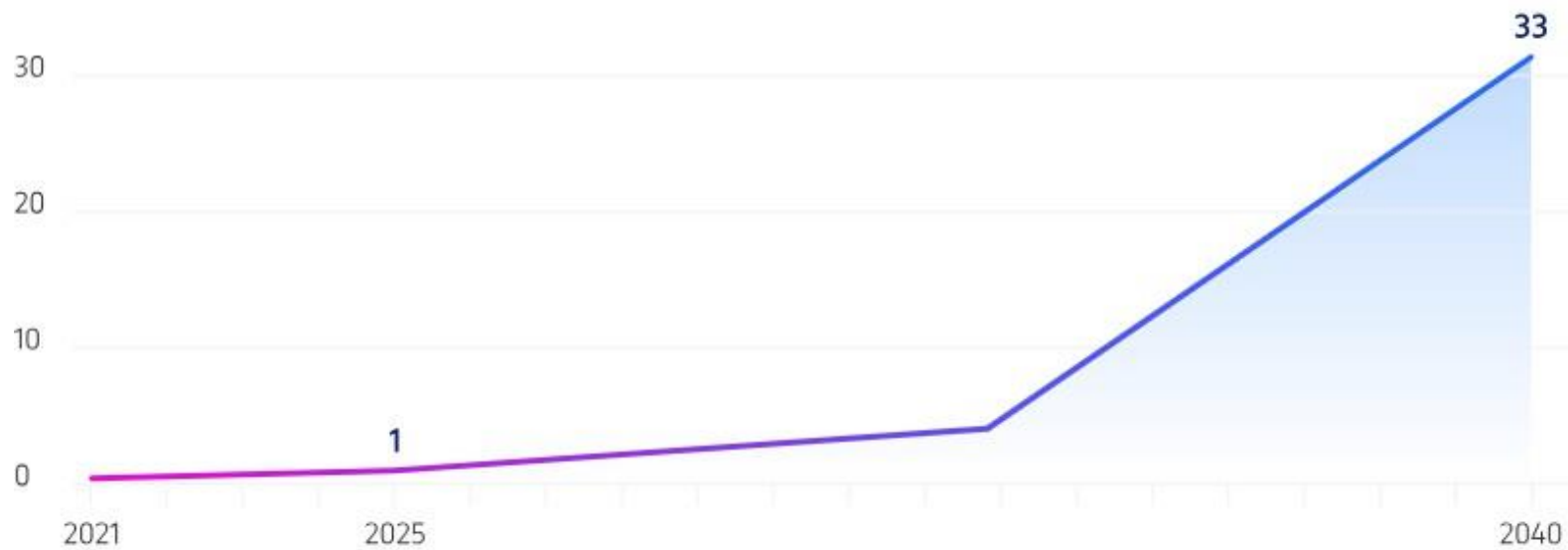
- **Smart cities.** Smart technology such as sensors, lights and meters are used to collect data that is designed to help improve operational efficiency, drive economic growth and improve quality of life for residents.
- **Smart Home:** Smart appliances learn through human interaction and response. AIoT appliances can also store and learn from user data to understand user habits to provide customized support.
- **Autonomous vehicles.** [Autonomous vehicles](#) rely on multiple video camera and sensor systems to gather data about nearby vehicles, monitor driving conditions and look for pedestrians.



Waymo(Google self driving car)

- Waymo is designed to detect pedestrians, Cyclists, vehicles on the road from distance of two football fields away in all directions
- Biggest challenge of big data as the computer has to process more than 1 million data per second

Annual number of sales in millions



Data source: deloitte.com—Autonomous Driving. Moonshot Project with Quantum Leap from Hardware to Software & AI Focus

AI in Automotive

Manufacturing

Modern vehicle manufacturers extensively rely on AI to design smarter cars, make car assembly more efficient using robots and exoskeletons, and streamline supply chain management.

Transportation

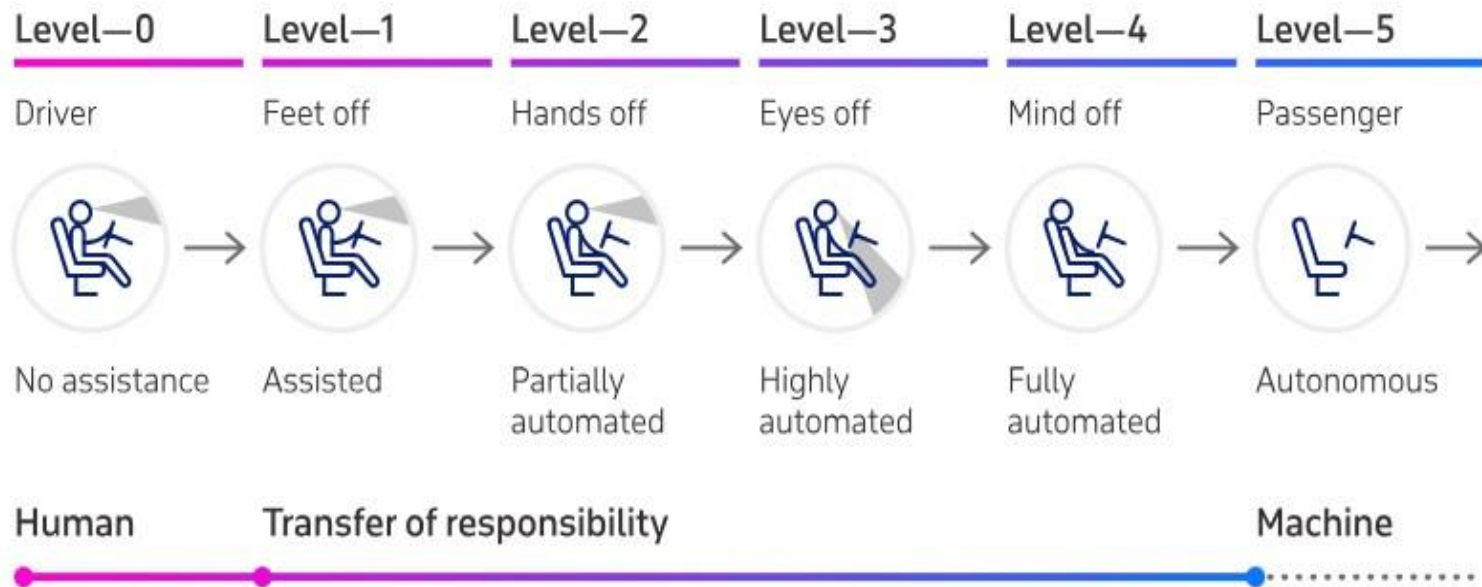
AI is the backbone of driver assistance technology. From helping drivers to navigate highways to recognizing their tiredness to detecting crucial defects in a vehicle, AI makes driving less stressful and more secure.

Service

By embedding IoT sensors into vehicles, sophisticated AI systems can predict engine failure and optimize battery performance based on how drivers use the car. On top of that, there is a growing demand for AI-enabled insurance, which can streamline claims settlements.

AI in driver Assistance

AI-assisted driving autonomy levels



- Experts claim that a vehicle needs to process more than 1 TB of data per second on average to become fully autonomous. Companies are turning to various cloud solutions, where the data is labeled, processed, and used to optimize these algorithms. However, the mass deployment of autonomous vehicles doesn't align with the current trend of low energy consumption and poses certain economic hurdles.

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LIDAR Sensors

- Lidar sensors gauge the distance between vehicles and other surrounding objects to command the car to slow down or speed up. Interestingly, Tesla, an obvious leader in the autonomous car manufacturing industry, is against the use of Lidar because it can't provide AI systems with sufficient information

- The autonomous vehicle is projected to contain more lines of code than any other software platform that has been created to date. By 2020, the typical vehicle is expected to contain more than 300 million lines of code and will contain more than 1 TB (terabytes) of storage and will require memory bandwidth of more than 1 TB per second to support the compute performance necessary for autonomous driving platforms.

What is the internet of medical things (IoMT) or healthcare IoT?

- The internet of medical things (IoMT) is the collection of medical devices and applications that connect to healthcare information technology systems through online computer networks. Medical devices equipped with Wi-Fi enable the [machine-to-machine](#) communication that is the basis of IoMT.
- Examples: Using remote patient monitoring (RPM) for people with chronic diseases and long-term conditions.
- Tracking patient medication orders.
- Tracking the location of patients admitted to hospitals.
- Collecting data from patients' wearable mobile health devices.
- Connecting ambulances en route to medical facilities to healthcare professionals.



Mobile & Ubiquitous Computing

Outline

- Background about Mobile & Ubiquitous Computing
- What are Pervasive or Ubiquitous Computing?
- Are they the same? Or are they different?
- What are some typical applications?
- What do you think about them? Future trend?
- What are the major obstacles?
- Summary



Mobile & Ubiquitous Computing

Background about Mobile & Ubiquitous Computing

- Computing (Data Processing)
 - Traditional Computation
 - Storage, Retrieval, and Processing of Data
- Location-aware Computing
 - Results of the computation depends on the location
 - Mobile Computing
- Ubiquitous Computing (5A Model)
 - Any time, Any where, Any Device, Any Data, Any Network
 - The Reverse of Virtual Reality
- Ubiquitous Intelligent Community
 - Network-centric (Connectivity)
 - Data-centric (Information Retrieval)
 - Smart Space/Smart World



Mobile & Ubiquitous Computing

What are Pervasive and Ubiquitous Computing?

From the Dictionary

Pervasive

[adjective] spread throughout

Ubiquitous

[adjective] being present every- where at once

Recursive Definitions

Pervasive computing is the trend towards increasingly ubiquitous

Ubiquitous computing is the trend towards increasingly pervasive



Mobile & Ubiquitous Computing

What are Pervasive and Ubiquitous Computing?

- Pervasive computing can be characterized as creating environments saturated with computing and wireless communication, and yet gracefully integrated with human users.
- Pervasive means “diffused throughout every part of”
- Ubiquitous means “everywhere”



Internet Tomorrow

Embodied Virtuality (Pervasive Computing)

The most profound technologies are those that disappear.

They weave themselves into the fabric of everyday life until they are indistinguishable from it.

Ubiquitous, Invisible Computing:

The process of drawing computers out of electronics shells

The virtuality of computer-readable data is **brought into the physical world.**



(Siemens)

Ubiquitous computing reside in the human world and poses no barrier to personal interactions

When almost every object contains a computer [..], obtaining information will be trivial

(M. Weiser (PARC), The Computer for the 21st Century, 1991)



Mobile & Ubiquitous Computing

What are Pervasive and Ubiquitous Computing?

- The term “ubiquitous” was coined by Mark Weiser in 1988
- Ubiquitous computing is the method of enhancing computer use by making many computers available throughout the physical environment, but making them effectively invisible to the user.



-Mark Weiser
Chief technologist at Xerox PARC
Father of "ubiquitous computing"
(1952-1999)

Reference: Mark Weiser, “The Computer for the 21st Century”, Scientific America, Sep. 1991



Mobile & Ubiquitous Computing

What are Pervasive and Ubiquitous Computing?

- Pervasive and Ubiquitous Computing means many computing devices serving people in every aspects of daily life without user's awareness of the existing devices.
- Pervasive and Ubiquitous computing is a very difficult integration of human factors, computer science, engineering, and social sciences.
- Some Keywords
 - Invisible
 - Transparent
 - Embedded
 - Mobile
 - Everywhere



Mobile & Ubiquitous Computing

What are Pervasive and Ubiquitous Computing?

- The 5As Model for Pervasive and Ubiquitous Computing
 - Anytime/Anywhere: 7 days x 24 hours, global access.
 - Any device: PC, Laptop, Netbook, PDA/smartphone.
 - Any network: Mobile/Wireless
 - Any data: e-mail, MMS message.
- Scenario

Sending and receiving **any data** through **any network** using **any device** at **any time** and **any where**.



Mobile & Ubiquitous Computing

Are they the same or are they different?

- In terms of concepts, they are the same.
- However, in the academic community:
 - Pervasive Computing:
 - Merging Existing Technologies
 - Mobile / Wireless Network
 - Embedded Systems
 - Sensor Networks
 - Wearable Computers
 - Ubiquitous Computing:
 - Emphasis on the Concepts
 - User-Oriented
 - Special Devices
 - Embedded / Device Computing



Mobile & Ubiquitous Computing

Major Trends in Computing

The Major Trends in Computing	
Mainframe	Many people share a computer
Personal Computer	One computer, one person
Internet, Distributed Computing	Transitional State
Ubiquitous Computing	Each of us making use of many computers



Mobile & Ubiquitous Computing

Some Applications and Future Trend of Computing?

- Smart Home
- Smart Electric Appliances / Devices
 - Refrigerator, Washing Machines, Microwave
 - Vending Machines
 - Mac notebook – sensing sound and light
 - PDA+Phone, iPod
 - SmartCards
- Location Based Services
 - Finding people and friends
 - Locating food and entertainment
 - Location-aware computing

Do you think this is the future trend of computing?



Mobile & Ubiquitous Computing

Some Applications and Future Trend of Computing?

Features of devices

- application-specific integrated circuitry
- Speech recognition
- Gesture recognition
- System on chip
- Perceptive interface
- Smart matter
- Reconfigurable processors
- Flexible transistors
- Field programmable logic gates
- Micro-electromechanical systems





Mobile & Ubiquitous Computing

Some Applications and Future Trend of Computing?

Eye Tap is a device that is worn in front of the eye

- Acts as a camera to record the scene available to the eye
- Acts as a display to superimpose a Computer-generated imagery on the original scene available to the eye
- In order to capture what the eye is seeing as accurately. The camera digitizes the reflected image of the scene and sends it to a computer. It processes the image and then sends it to a projector. The projector send the image to the other side of the beam splitter so that this computer generated image is reflected into the eye to be superimposed on the original scene.





Mobile & Ubiquitous Computing

Some Applications and Future Trend of Computing?

Fridges

- Tilting, pull-out 15.1" touch-screen for accessing all services
- Built-in stereo speakers, CCD camera and microphone for Entertainment, interactive and messaging services
- Information
 - Electronic calendar for keeping important dates
 - Electronic nutritional fact file for tips and information on products purchased
 - Track foods and their storage time in your fridge freezer-- This fridge can track the storage time of the food by ubiquitous computing
 - Electronic user features and maintenance manuals
 - Self diagnostic system for highlighting faults
 - Phone Number Management
 - External Management
 - Cooking Recipes
 - Weather Information
 - Handwriting Recognition





Mobile & Ubiquitous Computing

Some Applications and Future Trend of Computing?

Wearable Computer





Mobile & Ubiquitous Computing

Some Applications and Future Trend of Computing?

Implantable RFID chips

- Originally designed for animal tagging but are now being used in humans as well.
- An early experiment with RFID implants was conducted by British professor of cybernetics Kevin Warwick, who implanted a chip in his arm in 1998.
- Applied Digital Solutions proposes their chip's "unique under-the-skin format" as a solution to identity fraud, secure building access, computer access, storage of medical records, anti-kidnapping initiatives and a variety of law-enforcement applications.
- Combined with sensors to monitor body functions, the Digital Angel device could provide monitoring for patients.





Mobile & Ubiquitous Computing

Ubiquitous Computing Research Centre at HKBU

Collaborating Institutions

- Hong Kong Baptist University (HKBU) - Lead Institution
- Hong Kong University of Science and Technology (HKUST)
- Hong Kong Polytechnic University (PolyU)

Research Areas

1. Location Aware Computing;
2. RFID / Sensor Network;
3. Middleware/Software Development;
4. Mobile Data Management;
5. Ad Hoc/IPv6 Network;
6. Mobile Agents/Mobile Services;

Webpage for RCUC: <http://www.comp.hkbu.edu.hk/RCUC>



Mobile & Ubiquitous Computing

Geo-caching (Mobile Treasure Hunting)

A location-aware treasure hunting competition utilizing GPS, UMTS, Wi-Fi and PAN positioning technologies. This is one of the events in both 2008 & 2009 World Telecommunication and Information Society Day - Hong Kong





Mobile & Ubiquitous Computing

Services & Applications

- Goods Transport and Logistic
- Fleet Management
- Field Worker Deployment
- Resource Management & Warehouse Management
- Elderly, Blind & Mentally Retarded
- Tourism
- Infotainment
- Personal Safety & Security
- Education
- Library & Museum

Google Glass telehealth

- <https://www.youtube.com/watch?v=ssldTFWBv3E>



Mobile & Ubiquitous Computing

Major Obstacles

- Hardware
 - Size of the Device
 - Cost of the Device
 - Battery Life Time
- Software
 - Software Engineering Process
 - Languages
 - Interoperability
- Network Infrastructure
 - Scalability
 - Hybrid Network (Wired, Wireless, and Mobile)
 - Load balancing / Fault Tolerance
- Security
- Human-Computer Interface
- Privacy
- Human Factors / Social Issues



Mobile & Ubiquitous Computing

Summary

- Mobile computing is a reality and playing a major role in computing and communications.
- Certainly, Pervasive/Ubiquitous computing is a nice thing to have.
- It has a high impact on the future trend of computing.
- Many universities and researchers are now paying much more effort on its development.
- Pervasive/Ubiquitous computing is a very difficult integration of human factors, computer science, engineering, and social sciences.
- The human factors are the major obstacles to the adoption of Pervasive and Ubiquitous computing to our daily life

UNIT-4

IOT Networking

Connectivity Terminologies

IoT Node : These are machines , things or computers Connected to other nodes inside a LAN via the IoT LAN, May be sometimes connected to the internet through a WAN directly

IoT LAN : It is Local, Short range Comm, May or may not connect to Internet, Building or Organization wide

IoT WAN: Connection of various network segments, Organizationally and geographically wide, Connects to the internet

IoT Gateway : A router connecting the IoT LAN to a WAN to the Internet, Can implement several LAN and WAN, Forwards packets between LAN and WAN on the IP layer

IoT Proxy: Performs active application layer functions between IoT nodes and other entities

Gateway Prefix Allotment:

- ✓ One of the strategies of address conservation in IoT is to use local addresses which exist uniquely within the domain of the gateway. These are represented by the circles in this slide.
- ✓ The network connected to the internet has routers with their set of addresses and ranges.
- ✓ These routers have multiple gateways connected to them which can forward packets from the nodes, to the Internet, only via these routers. These routers assign prefixes to gateways under them, so that the gateways can be identified with them.

Impact of Mobility on Addressing

- ✓ The network prefix changes from 1 to 2 due to movement, making the IoT LAN safe from changes due to movements.
- ✓ IoT gateway WAN address changes without change in LAN address. This is achieved using ULA.
- ✓ The gateways assigned with prefixes, which are attached to a remote anchor point by using various protocols such as Mobile IPv6, and are immune to changes of network prefixes.
- ✓ This is achieved using LU. The address of the nodes within the gateways remain unchanged as the gateways provide them with locally unique address and the change in gateway's network prefix doesn't affect them.
- ✓ Sometimes, there is a need for the nodes to communicate directly to the internet. This is achieved by tunneling, where the nodes communicate to a remote anchor point instead of channeling their packets through the router which is achieved by using tunneling protocols such as IKEv2: internet key exchange version 2

Multihoming

Multihoming is the practice of connecting a host or a computer network to more than one network. This can be done in order to increase reliability or performance or to reduce cost. There are several different ways to perform multihoming.

Host multihoming

A single host may be connected to multiple networks. For example, a mobile phone might be simultaneously connected to a WiFi network and a 3G network, and a desktop computer might be connected to both a home network and a VPN. A multihomed host usually is assigned multiple addresses, one per connected network.

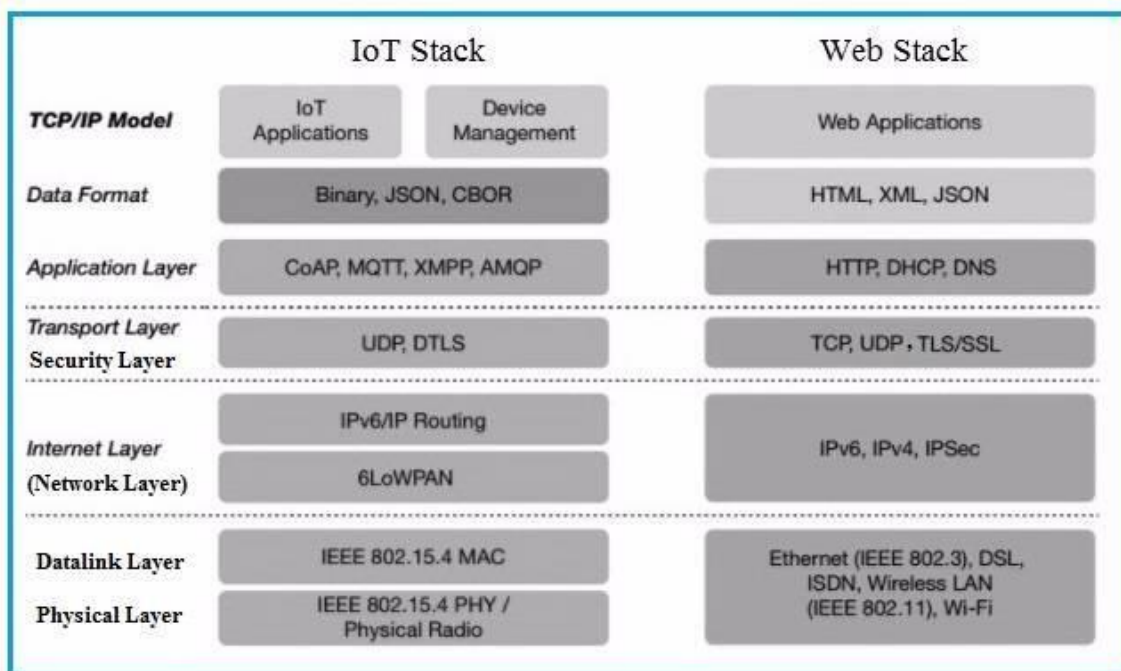
Classical multihoming

In classical multihoming a network is connected to multiple providers, and uses its own range of addresses (typically from a Provider Independent (PI) range). The network's edge routers communicate with the providers using a dynamic routing protocol, typically BGP, which announces the network's address range to all providers. If one of the links fails, the dynamic routing protocol recognizes the failure within seconds or minutes, and reconfigures its routing tables to use the remaining links, transparently to the hosts.

Multihoming with multiple addresses

In this approach, the network is connected to multiple providers, and assigned multiple address ranges, one for each provider. Hosts are assigned multiple addresses, one for each provider.

Deviation from regular Web



Features	IoT Stack	Web Stack
Function or application	It is used in constrained network having low power, low bandwidth and low memory requirements.	It is used in non-constrained network having no limits on power/BW/memory.
Size of data to be transported	tens of bytes	hundreds or thousands of bytes
Data format	It uses CBOR (Concise Binary Object Representation) format as IoT is used for tiny messages. CBOR is based on JSON though CBOR uses binary encoding while JSON uses text encoding.	It uses HTML, XML and JSON formats.
Application Layer	It uses CoAP protocol at application layer.	It uses HTTP protocol at application layer.
Transport layer	It uses UDP which is faster due to smaller header size compare to TCP. It is lighter protocol compare to TCP.	It uses TCP which is connection oriented and slower compare to UDP.
Security layer	It uses DTLS (Datagram Transport Layer Security) protocol for security.	It uses TLS/SSL protocols for the same.
Internet layer	It uses 6LoWPAN to convert large IPv6 packets into small size packets to be carried on wireless medium as per bluetooth, zigbee etc. standards. It does fragmentation and reassembly. It also does header compression to reduce packet size.	It does not require protocols like 6LoWPAN. Fragmentation and reassembly is taken care by transport layer (i.e. TCP) itself.
Datalink or MAC layer	It will have MAC layer as per IoT wireless technology used viz. bluetooth, zigbee, zwave etc. It takes care of medium access control and resource allocation and management.	It will have MAC layer as per LAN or WLAN or DSL or ISDN technologies.
Physical layer and Radio Frequency (RF) layer	It will have physical layer (baseband) as per IoT wireless technologies viz. bluetooth, zigbee, zwave etc. It uses frequencies as per cellular or indoor wireless technologies and country wide allocations for the same.	It will have PHY layer as per LAN or WLAN or DSL or ISDN technologies.

IoT identification and Data protocols

IPv4:

IP version four addresses are 32-bit integers which will be expressed in dotted decimal notation. Example- 192.0.2.126 could be an IPv4 address.

Characteristics of IPv4

- IPv4 could be a 32-Bit IP Address.
- IPv4 could be a numeric address, and its bits are separated by a dot.
- The number of header fields is twelve and the length of the header field is twenty.
- It has Unicast, broadcast, and multicast style of addresses.
- IPv4 supports VLSM (Virtual Length Subnet Mask).
- IPv4 uses the Post Address Resolution Protocol to map to the MAC address.
- RIP may be a routing protocol supported by the routed daemon.
- Networks ought to be designed either manually or with DHCP.
- Packet fragmentation permits from routers and causing host.

IPv4 Datagram Header

0	4	8	16	19	24	31
Version	IHL	Type of Service	Total Length			
Identification			Flags	Fragment Offset		
TTL		Protocol	Header Checksum			
Source IP Address						
Destination IP Address						
Options					Padding	

Fig: IPv4 Frame Format

Version:

This field indicates the version number of the IP packet so that the revised version can be distinguished from the previous version. The current IP version is 4.

Internet Header Length (IHL):

It specifies the length of the IP header in unit 32 bits. In case of no option present in the IP header, IHL will have a value of 5. So, if the value of IHL is more than 5 then the length of the option field can be easily calculated.

Type of Service: This field specifies the priority of the packets based on delay, throughput, reliability and cost requirements. Three bits are assigned for priority level and four bits for specific requirements (delay, throughput, reliability and cost).

Total Length:

This field specifies the number of bytes of the IP packet including header and data. As 16 bits are assigned to this field, the maximum length of the packet is 65535 bytes.

Identification:

The identification field is used to identify which packet a particular fragment belongs to so that fragments for different packets don't get mixed up.

Flags:

The flag field has three bits:

1. Unused bit
2. Don't fragment (DF) bit
3. More fragment (MF) bit

Fragment Offset:

The fragment offset field identifies the location of the fragment in a packet. The value measures the offset in a unit of 8 bytes, between the beginning of the packet to be fragmented and the beginning of the fragment.

Time to live (TTL):

This field is used to indicate the amount of time in seconds a packet is allowed to remain in the network.

Protocol:

This field specifies the protocol that is to receive the IP data at the destination host.

Header Checksum:

This field verifies the integrity of the header of the IP packet. The integrity of the data part is left to the upper layer protocols. The checksum is generated by the source and it is sent along with the frame header to the next router.

Source IP address & Destination IP address:

These two fields contain the IP addresses of the source and destination hosts respectively.

Options:

Options fields are rarely used to include special features such as security level, the route to be taken and time stamp at each router. It is used in RSVP.

Padding:

This field is used to make the header a multiple of 32-bit words.

IPv6

Internet Protocol version 6 (IPv6) is also known as **Internet Protocol next generation (IPng)**. It also accommodates more feature to meet the global requirement of growing Internet.

To allocate a sufficient number of network address, IPv6 allows 128 bits of IP address separated into 8 sections of 2 bytes each. Unlike IPv4 where the address is represented as dotted-decimal notation, IPv6 uses hexadecimal numbers and colon (":") is used as a delimiter between the sections.

Example: IPv6 address may be like this :

FA20:B120:6230:0000:0000:CE12:0006:ABDF

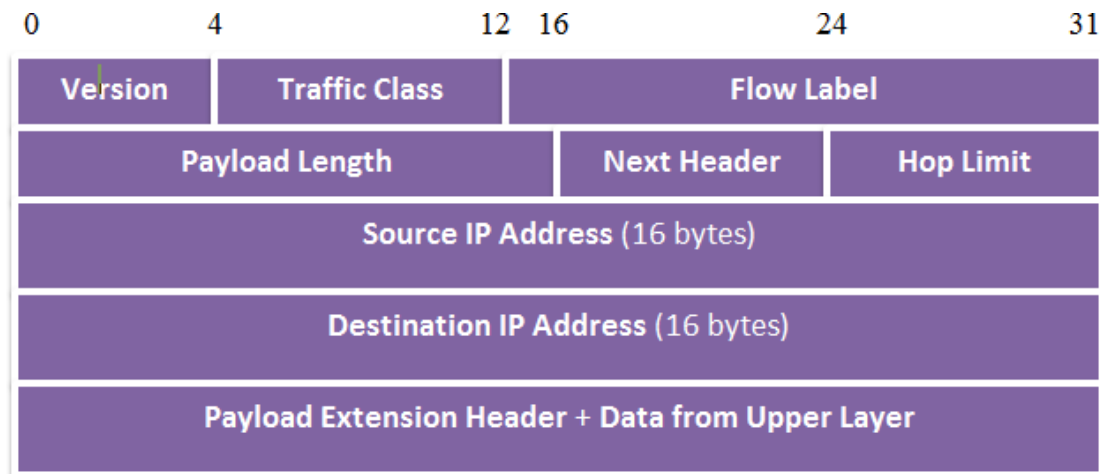


Fig: IPv6 Packet Format

Version: This field is 4 bits long and it defines the version of the IP packet. The value of it for IPv6 is 6 and IPv4 its value is 4. During the transition period from IPv4 to IPv6, the routers will be able to distinguish the two versions of the IP packets.

Traffic Class: This field is 20 bits long and it is used to distinguish between the different requirements for real-time delivery services.

Flow Label: This field is 20 bits long and it is used to allow the source and destination nodes to set up a pseudo connection with particular properties and requirements. It is designed to provide special handling of a particular flow of data.

Payload Length: It is of 2 bytes length and signifies the number of bytes that follow the 40 bytes base header. It is the length of the IP datagram excluding the base header.

Next Header: This field is of 1 byte length and it defines one of the extension headers that follow the base header. The extension headers also contain this field to indicate the next header. If this is the last IP header then Next header field tells which of the transport protocols (TCP or UDP) the packet is to be passed.

Hop Limit: This field contains 1 byte and it signifies the maximum number of hops a packet can travel. The time to live field in the IPv4 header did the same task, except that in IPv4 it was counted in time and in IPv6 it is counted in terms of the number of routers.

Source Address: It is 16 bytes long and contains the IP address of the source machine to the network interface.

Destination Address: It is 16 bytes long and usually contains the IP address of the ultimate destination machine to the network interface. In case of specific routing, it may contain the IP address of the next router.

Extension Header: Some of the fields IPv4 that are missing in IPv6 is necessary in some of the cases. To handle this problem, IPv6 has introduced the concept of the extension header. There are one or more of the six possible extension headers. These headers appear directly after the base header.

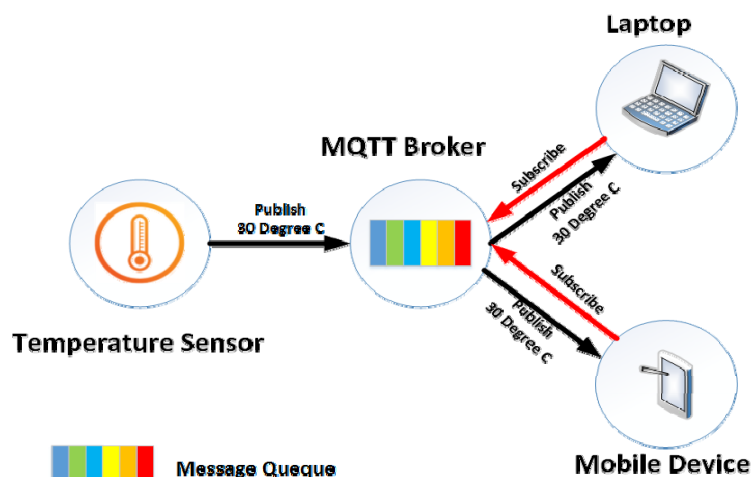
MQTT

- ✓ It is a publish-subscribe-based lightweight messaging protocol for use in conjunction with the TCP/IP protocol.
- ✓ Designed to provide connectivity (mostly embedded) between applications and middlewares on one side and networks and communications on the other side.
- ✓ A message broker controls the publish-subscribe messaging pattern.
- ✓ A topic to which a client is subscribed is updated in the form of messages and distributed by the message broker.
- ✓ Designed for: Remote connections, Limited bandwidth, Small-code footprint

MQTT Components

- **Publishers:** Lightweight sensors
- **Subscribers:** Applications interested in sensor data
- **Brokers:** Connect publishers and subscribers and Classify sensor data into topics

Communication:



- ✓ The protocol uses a **publish/subscribe** architecture (HTTP uses a request/response paradigm).
- ✓ Publish/subscribe is **event-driven** and enables messages to be pushed to clients.
- ✓ The central **communication point is the MQTT broker**, which is in charge of dispatching all messages between the senders and the rightful receivers.
- ✓ Each client that publishes a message to the broker, includes a **topic** into the message. The **topic is the routing information for the broker**.
- ✓ Each client that wants to receive messages subscribes to a certain topic and the broker delivers all messages with the matching topic to the client.
- ✓ Therefore the clients don't have to know each other. They only communicate over the topic.

- ✓ This architecture enables highly scalable solutions without dependencies between the data producers and the data consumers.

Applications

- ✓ **Facebook Messenger** uses MQTT for online chat.
- ✓ **Amazon Web Services** use Amazon IoT with MQTT.
- ✓ **Microsoft Azure** IoT Hub uses MQTT as its main protocol for telemetry messages.
- ✓ The **EVERYTHING IoT platform** uses MQTT as an M2M protocol for millions of connected products.
- ✓ **Adafruit** launched a free MQTT cloud service for IoT experimenters called Adafruit IO.

SMQTT

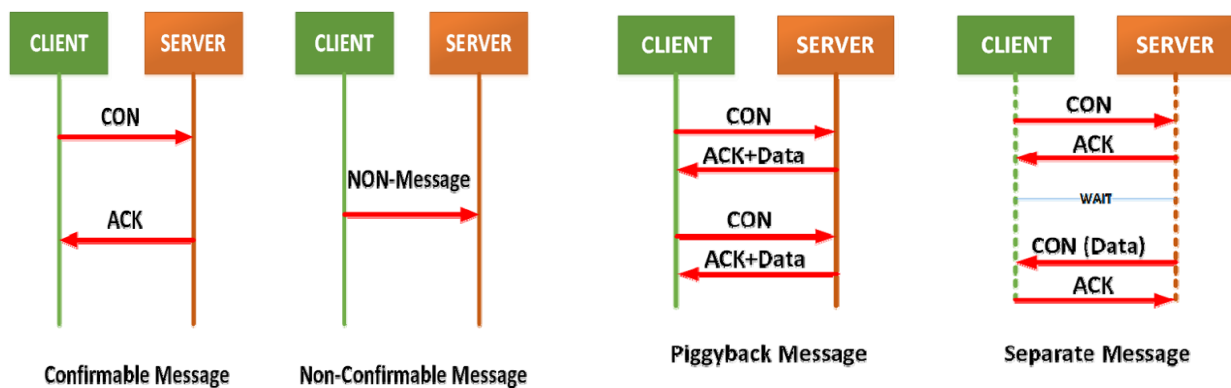
- ✓ **Secure MQTT** is an extension of MQTT which uses encryption based on lightweight attribute based encryption.
- ✓ The main advantage of using such encryption is the broadcast encryption feature, in which one message is encrypted and delivered to multiple other nodes, which is quite common in IoT applications.
- ✓ In general, the algorithm consists of four main stages: setup, encryption, publish and decryption.

CoAP

- ✓ CoAP – **Constrained Application Protocol**.
- ✓ **Web transfer protocol** for use with constrained nodes and networks.
- ✓ **Designed for Machine to Machine** (M2M) applications such as smart energy and building automation and Based on **Request-Response model** between end-points
- ✓ Client-Server interaction is **asynchronous over a datagram oriented transport protocol** such as UDP
- ✓ The Constrained Application Protocol (CoAP) is a session layer protocol designed by IETF Constrained RESTful Environment (CoRE) working group to provide lightweight RESTful (HTTP) interface.
- ✓ Representational State Transfer (REST) is the standard interface between HTTP client and servers.
- ✓ Lightweight applications such as those in IoT, could result in significant overhead and power consumption by REST.
- ✓ CoAP is designed to enable low-power sensors to use RESTful services while meeting their power constraints
- ✓ Built over UDP, instead of TCP (which is commonly used with HTTP) and has a light mechanism to provide reliability.
- ✓ CoAP architecture is divided into two main sub-layers:

- Messaging
 - Request/response.
- ✓ The messaging sub-layer is responsible for reliability and duplication of messages, while the request/response sub-layer is responsible for communication.
- ✓ CoAP has four messaging modes:
- Confirmable
 - Non-confirmable
 - Piggyback
 - Separate

CoAP Request-Response Model



- ✓ Confirmable and non-confirmable modes represent the reliable and unreliable transmissions, respectively, while the other modes are used for request/response.
- ✓ Piggyback is used for client/server direct communication where the server sends its response directly after receiving the message, i.e., within the acknowledgment message.
- ✓ On the other hand, the separate mode is used when the server response comes in a message separate from the acknowledgment, and may take some time to be sent by the server.
- ✓ Similar to HTTP, CoAP utilizes GET, PUT, PUSH, DELETE messages requests to retrieve, create, update, and delete, respectively

XMPP

- ✓ XMPP – Extensible Messaging and Presence Protocol.
- ✓ A communication protocol for message-oriented middleware based on XML (Extensible Markup Language).
- ✓ Real-time exchange of structured data.
- ✓ It is an open standard protocol
- ✓ XMPP uses a client-server architecture.

- ✓ As the model is decentralized, no central server is required.
- ✓ XMPP provides for the discovery of services residing locally or across a network, and the availability information of these services.
- ✓ Well-suited for cloud computing where virtual machines, networks, and firewalls would otherwise present obstacles to alternative service discovery and presence-based solutions.
- ✓ Open means to support machine-to-machine or peer-to-peer communications across a diverse set of networks.

Applications:

- ✓ Publish-subscribe systems
- ✓ Signaling for VoIP
- ✓ Video
- ✓ File transfer
- ✓ Gaming
- ✓ Internet of Things applications: Smart grid and Social networking services

AMQP

- ✓ Advanced Message Queuing Protocol.
- ✓ Open standard for passing business messages between applications or organizations.
- ✓ Connects between systems and business processes.
- ✓ It is a binary application layer protocol.
- ✓ Basic unit of data is a *frame*

Components

Exchange:

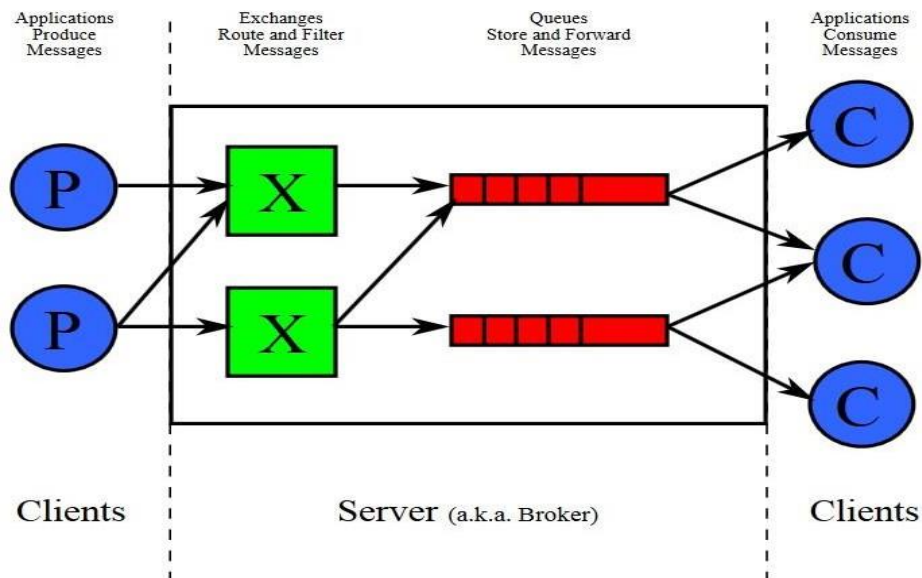
- ✓ Part of Broker
- ✓ Receives messages and routes them to Queues

Queue :

- ✓ Separate queues for separate business processes
- ✓ Consumers receive messages from queues

Bindings:

- ✓ Rules for distributing messages (who can access what message, destination of the message)



AMQP Features

- ✓ Targeted QoS (Selectively offering QoS to links)
- ✓ Persistence (Message delivery guarantees)
- ✓ Delivery of messages to multiple consumers
- ✓ Possibility of ensuring multiple consumption
- ✓ Possibility of preventing multiple consumption
- ✓ High speed protocol

Applications

- ✓ Monitoring and global update sharing.
- ✓ Connecting different systems and processes to talk to each other.
- ✓ Allowing servers to respond to immediate requests quickly and delegate time consuming tasks for later processing.
- ✓ Distributing a message to multiple recipients for consumption.
- ✓ Enabling offline clients to fetch data at a later time.
- ✓ Introducing fully asynchronous functionality for systems.
- ✓ Increasing reliability and uptime of application deployments.

Unit – 5

SMART WORLD

1. SMART WORLD

Introduction: Sensing & actuation, Communication, Electronics in Smart city, 5G Technology, Communication protocols, Integration of Sensors in Robots and Artificial Intelligence, Human-Machine Interaction, Industrial IoT- Application Domains: Healthcare, Power Plants, Inventory Management & Quality Control, Plant Safety and Security (Including AR and VR safety applications), Facility Management., Intellectual Property Rights- Case Studies - Milk Processing and Packaging Industries.

1.1 Introduction: Sensing & actuation

Sensors turn a physical input into an electrical output, and actuators do the opposite. They take electrical signals from control modules and turn them into physical outputs. In a typical IoT system, a sensor may collect information and route to a control center. There, previously defined logic dictates the decision. As a result, a corresponding command controls an actuator in response to that sensed input. Thus, sensors and actuators in IoT work together from opposite ends. IoT sensor – An IoT device that observes one or more properties of a physical entity and converts those properties into information. IoT actuator – An IoT device that can change one or more properties of a physical entity in response to received information. An actuator is a device that produces a motion by converting energy and signals going into the system. The motion it produces can be either rotary or linear. An actuator is a device that produces a motion by converting energy and signals going into the system. Sensors play an important role in creating solutions using IoT. Sensors are devices that detect external information, replacing it with a signal that humans and machines can distinguish. An actuator is a part of a device or machine that helps it to achieve physical movements by converting energy, often electrical, air, or hydraulic, into mechanical force. Common examples of actuators include electric motors, stepper motors, jackscrews, electric muscular stimulators in robots, etc.

A sensor is a device that measures physical input from its environment and converts it into data that can be interpreted by either a human or a machine. People use sensors to measure temperature, gauge distance, detect smoke, regulate pressure and a myriad of other uses. A sensor is a device that detects the change in the environment and responds to some output on the other system. A sensor converts a physical phenomenon into a measurable analog voltage (or sometimes a digital signal) converted into a human-readable display or transmitted for

reading or further processing. The most frequently used different types of sensors are classified based on the quantities such as Electric current or Potential or Magnetic or Radio sensors, Humidity sensor, Fluid velocity or Flow sensors, Pressure sensors, Thermal or Heat or Temperature sensors, Proximity sensors, Optical sensors, Position sensors.

1.1.1 Communication

The sensor nodes can communicate among themselves using radio signals. A wireless sensor node is equipped with sensing and computing devices, radio transceivers and power components. The working mode of the sensor nodes may be either continuous or event driven. A Wireless sensor network can be defined as a network of devices that can communicate the information gathered from a monitored field through wireless links. The data is forwarded through multiple nodes, and with a gateway, the data is connected to other networks like wireless Ethernet. A wireless sensor-actuator network (WSAN) is composed of sensor and actuator nodes interconnected in a wireless channel. Sensor nodes can deliver messages to only nearer nodes due to weak radio and messages are forwarded by sensor nodes to an actuator node. The sensor network connects to the internet or computer networks to transfer data for analysis and use. Sensor network nodes cooperatively sense and control the environment. They enable interaction between persons or computers and the surrounding environment.

They are hardware devices that produce a measurable response to a change in a physical condition like temperature or pressure. Sensors measure physical data of the parameter to be monitored and have specific characteristics such as accuracy, sensitivity etc. The continual analog signal produced by the sensors is digitized by an analog-to-digital converter and sent to controllers for further processing. Some sensors contain the necessary electronics to convert the raw signals into readings which can be retrieved via a digital link (e.g. I2C, SPI) and many convert to units such as °C. Most sensor nodes are small in size, consume little energy, operate in high volumetric densities, be autonomous and operate unattended, and be adaptive to the environment. As wireless sensor nodes are typically very small electronic devices, they can only be equipped with a limited power source of less than 0.5-2 ampere-hour and 1.2-3.7 volts.

Sensors are classified into three categories: passive, omnidirectional sensors; passive, narrow-beam sensors; and active sensors. Passive sensors sense the data without actually manipulating the environment by active probing. They are self powered; that is, energy is needed only to amplify their analog signal. Active sensors actively probe the environment, for example, a sonar or radar sensor, and they require continuous energy from a power source.

Narrow-beam sensors have a well-defined notion of direction of measurement, similar to a camera. Omnidirectional sensors have no notion of direction involved in their measurements.

Most theoretical work on WSNs assumes the use of passive, omnidirectional sensors. Each sensor node has a certain area of coverage for which it can reliably and accurately report the particular quantity that it is observing. Several sources of power consumption in sensors are: signal sampling and conversion of physical signals to electrical ones, signal conditioning, and analog-to-digital conversion. Spatial density of sensor nodes in the field may be as high as 20 nodes per cubic meter.

1.2 Electronics in Smart city

The smart city concept integrates information and communication technology (ICT), and various physical devices connected to the IoT network to optimize the efficiency of city operations and services and connect to citizens. Smart city applications are developed to manage urban flows and allow for real-time responses.

1.2.1 Smart city

A smart city is the one that uses information and communication technologies (ICT) to increase operational efficiency, share information with the public and improve both the quality of government services and citizen welfare. Smart cities use intelligent solutions to optimize infrastructure and smart and responsive governance to engage citizens in the management of their city. A system of sensors, networks, and applications collect useful data, like traffic congestion, energy use, and CO₂ levels.

A smart city is the one that uses information and communication technologies (ICT) to increase operational efficiency, share information with the public and improve both the quality of government services and citizen welfare.

Smart cities use intelligent solutions to optimize infrastructure and smart and responsive governance to engage citizens in the management of their city.

A system of sensors, networks, and applications collect useful data, like traffic congestion, energy use, and CO₂ levels.

The purpose of the Smart Cities Mission is to drive economic growth and improve the quality of life of people by enabling local area development and harnessing technology, especially technology that leads to Smart outcomes. Smart cities use Internet of Things (IoT)

devices such as connected sensors, lights, and meters to collect and analyze data. The cities then use this data to improve infrastructure, public utilities and services, and more. The four pillars of comprehensive development-institutional, physical, social and economic infrastructure.

1.2.2 Features of Smart Cities

- adequate water supply
- assured electricity supply
- sanitation, including solid waste management
- efficient urban mobility and public transport
- affordable housing, especially for the poor
- robust IT connectivity and digitalization
- good governance, especially e-Governance and citizen participation

1.2.3 Electronics in Smart city

- Smart lighting
- Charging stations
- Smart parking
- Smart Buildings

Smart lighting

In a smart city, the lighting system is integrated with advanced sensors and communication channels to obtain a Smart Lighting System (SLS). The goal of an SLS is to obtain an autonomous and more efficient lighting management system. Smart cities use IoT devices such as connected sensors, lights, and meters to collect and analyze data. The cities then use this data to improve infrastructure, public utilities and services, and more.

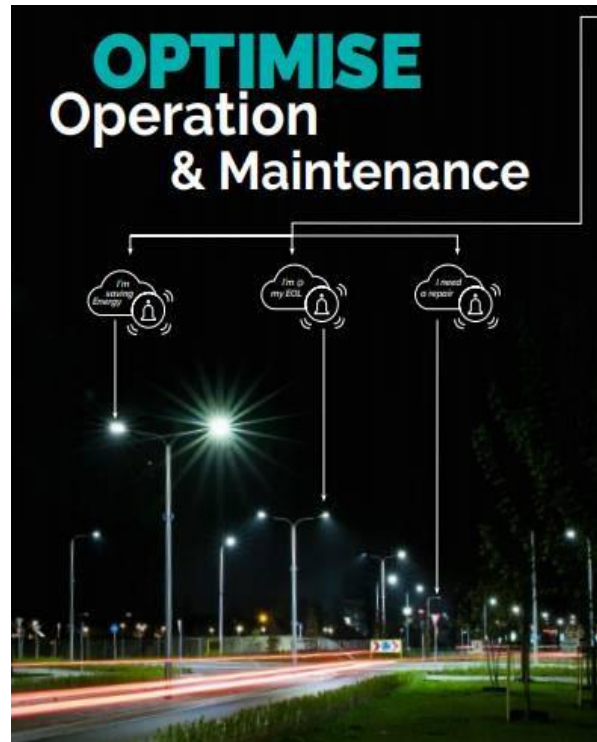


Figure 4.1 Smart Lighting

Minimise Environmental IMPACT OF YOUR CITY

The need for change is clear. Many reports all over the world support the fact that climate is changing. We have a joint responsibility to act and change the way we live. One of the key factors in this change is the use of energy. Apart from the change from fossil fuels to renewable energy, we urgently need to cut on the amount of energy being used.

With the expected growth of energy usage by 35% in 2030, the fact that lighting accounts for 29% of the total global usage* and that lighting accounts for 30-50% of an average city's energy bill*, there is a great opportunity to improve by adopting Smart Lighting.

*Source: www.bulb.org Energy Efficient Street Lighting report by Susceptor Expertise Center

Cut Energy Use

Our smart street lighting solutions allow energy savings of up to 80%. In absence of human presence, smart streetlights burn at a predefined level of brightness, for instance, at 20% of the full capacity. This stops unnecessary wastage of energy. When the presence of a pedestrian, cyclist, or a car is detected, the smart streetlights will adjust their brightness according to a pre-defined level. Other alternative is simply putting the lights to a pre-defined dimming schedule.

Lower CO₂ Emissions

Our intelligent street lighting solution significantly lowers CO₂ emissions. CO₂ or carbon dioxide occupies the largest share of today's greenhouse gases (the air pollution). This is bad for all living beings. Street lights waste a lot of energy, which contributes significantly to the increase in CO₂. Smart street lighting solutions from Twilight save energy radically, thereby lowering the CO₂ emissions considerably and creating a healthier, more sustainable living environment.

Curb Light Pollution

Our state-of-the-art street lighting solutions reduce light pollution. Regular streetlights keep on burning at their full brightness, even when there is no one around. This negatively affects nocturnal animals as well as human biology. Smart streetlights burn at a pre-defined level of brightness, for example, 20% of the full capacity, during off-peak hours. This feature lowers the unwanted light pollution and in turn offers a healthier environment, soothing ecosystem, and unspoiled aesthetic setting.

Figure 4.2 Smart Lighting impact

Smart parking

Smart Parking is a parking strategy that combines technology and human innovation in an effort to use as few resources as possible—such as fuel, time and space—to achieve faster, easier and denser parking of vehicles for the majority of time they remain idle. Smart Parking and its sister approach, Intelligent Transportation, are based on the fundamental ecological principle that we are all connected. Parking and transportation are both essential in the movement of people and goods. The Smart Parking and Intelligent Transportation vision and overlapping technologies are steadily melding into one integrated stream.



Figure 4.3 Smart Parking

Smart Buildings

A smart building is any structure that uses automated processes to automatically control the building's operations including heating, ventilation, air conditioning, lighting, security and other systems. A smart building uses sensors, actuators and microchips, in order to collect data and manage it according to a business' functions and services. This infrastructure helps owners, operators and facility managers improve asset reliability and performance, which reduces energy use, optimizes how space is used and minimizes the environmental impact of buildings.

Smart buildings use Internet of Things (IoT) devices—sensors, software, online connectivity—to monitor various building characteristics, analyze the data, and generate insights around usage patterns and trends that can be used to optimize the building's

environment and operations. Smart buildings use information technology during operation to connect a variety of subsystems, which typically operate independently, so that these systems can share information to optimize total building performance. Smart buildings look beyond the building equipment within their four walls.

Semiconductor Solutions for IoT-Enabled Smart Buildings

- Climate control, including temperature, humidity, vibration, etc.
- CO monitoring.
- Electrical usage.
- Fire detection.
- Heating, ventilation and air conditioning systems (HVAC)
- Lighting control.
- Occupancy.
- Predictive maintenance.



Figure 4.4 Smart Building

1.3 5G Technology

5G is the 5th generation mobile network. It is a new global wireless standard after 1G, 2G, 3G, and 4G networks. 5G enables a new kind of network that is designed to connect virtually everyone and everything together including machines, objects, and devices. 5G wireless technology is meant to deliver higher multi-Gbps peak data speeds, ultra low latency, more reliability, massive network capacity, increased availability, and a more uniform user experience to more users. Higher performance and improved efficiency empower new user experiences and connects new industries.

1.3.1 Evolution from 1G to 5G Technology

Generation	Speed	Technology	Key Features
1G (1970 –1980s)	14.4 Kbps	AMPS,NMT, TACS	Voice only services
2G (1990 to 2000)	9.6/ 14.4 Kbps	TDMA,CDMA	Voice and Data services
2.5G to 2.75G (2001-2004)	171.2 Kbps 20-40 Kbps	GPRS	Voice, Data and web mobile internet, low speed streaming services and email services.
3G (2004-2005)	3.1 Mbps 500- 700 Kbps	CDMA2000 (1xRTT, EVDO) UMTS and EDGE	Voice, Data, Multimedia, support for smart phone applications, faster web browsing, video calling and TV streaming.
3.5G (2006-2010)	14.4 Mbps 1- 3 Mbps	HSPA	All the services from 3G network with enhanced speed and more mobility.
4G (2010 onwards)	100-300 Mbps. 3-5 Mbps 100 Mbps (Wi-Fi)	WiMax, LTE and Wi-Fi	High speed, high quality voice over IP, HD multimedia streaming, 3D gaming, HD video conferencing and worldwide roaming.
5G (Expecting at the end of 2019)	1 to 10 Gbps	LTE advanced schemes, OMA and NOMA	Super fast mobile internet, low latency network for mission critical applications, Internet of Things, security and surveillance, HD multimedia streaming, autonomous driving, smart healthcare applications.

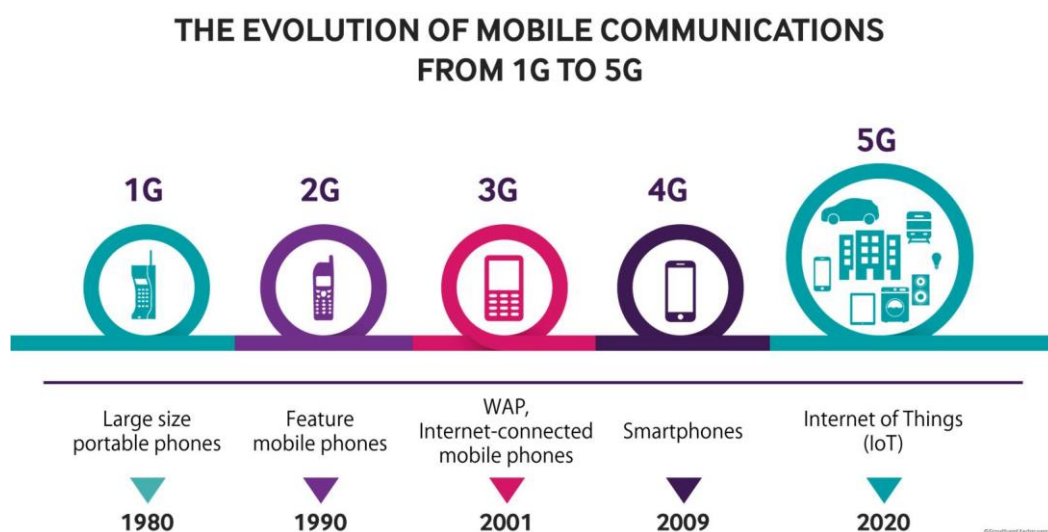


Figure 4.5 Evolution from 1G to 5G

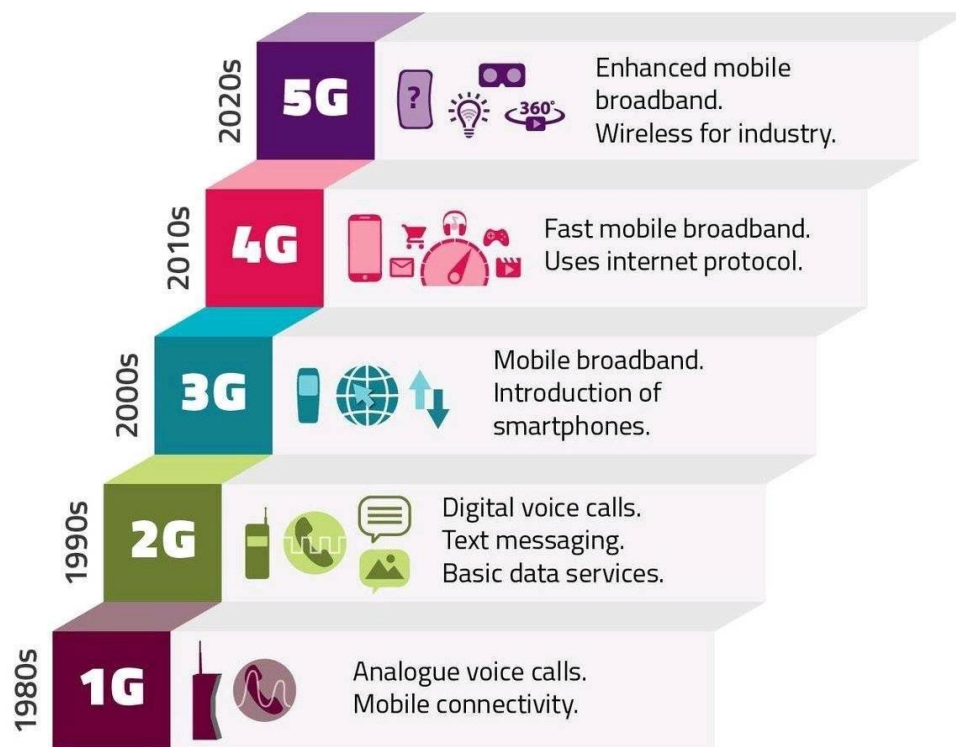


Figure 4.6 Transition from 1G to 5G Technology

1.3.2 4G – Fourth generation communication system

4G systems are enhanced version of 3G networks developed by IEEE, offers higher data rate and capable to handle more advanced multimedia services. LTE and LTE advanced wireless technology used in 4th generation systems. Furthermore, it has compatibility with previous version thus easier deployment and upgrade of LTE and LTE advanced networks are possible.

Simultaneous transmission of voice and data is possible with LTE system which significantly improve data rate. All services including voice services can be transmitted over IP packets.

Complex modulation schemes and carrier aggregation is used to multiply uplink / downlink capacity. Wireless transmission technologies like WiMax are introduced in 4G system to enhance data rate and network performance.

Key features of 4G system

- Much higher data rate up to 1Gbps
- Enhanced security and mobility
- Reduced latency for mission critical applications
- High definition video streaming and gaming
- Voice over LTE network VoLTE (use IP packets for voice)
- Disadvantages of 4G system

- Expensive hardware and infrastructure
- Costly spectrum (most countries, frequency bands are too expensive)
- High end mobile devices compatible with 4G technology required, which is costly
- Wide deployment and upgrade is time consuming

1.3.3 5G – Fifth generation communication system

5G will be using advanced technologies to deliver ultra fast internet and multimedia experience for customers. Current LTE advanced networks will transform into supercharged 5G networks in future. In order to achieve higher data rate, 5G technology will use millimeter waves and unlicensed spectrum for data transmission. Complex modulation technique has been developed to support massive data rate for Internet of Things. Cloud based network architecture will extend the functionalities and analytical capabilities for industries, autonomous driving, healthcare and security applications.

Key features of 5G technology

- Ultra fast mobile internet up to 10Gbps
- Low latency in milliseconds (significant for mission critical applications)
- Total cost deduction for data
- Higher security and reliable network
- Uses technologies like small cells, beam forming to improve efficiency
- Forward compatibility network offers further enhancements in future
- Cloud based infrastructure offers power efficiency, easy maintenance and upgrade of hardware

5G: Much More Than Just “Ultrafast Internet” on the Smartphone. 5G will do much more than significantly improve network connection. It provides new opportunities, enabling to deliver ground breaking solutions that reach across society. Imagine billions of connected devices gathering and sharing information in real time to reduce road accidents; or life-saving applications that can take flight thanks to lag-free guaranteed connections; or production lines so predictive they can prevent interruptions well before they occur.

5G runs on the same radio frequencies that are currently being used for your smartphone, on Wi-Fi networks and in satellite communications, but it enables technology to go a lot further.

Beyond being able to download a full-length HD movie to your phone in seconds (even from a crowded stadium), 5G is really about connecting things everywhere – reliably, without lag – so people can measure, understand and manage things in real time. This has enormous potential – and together, will take it to the next level.

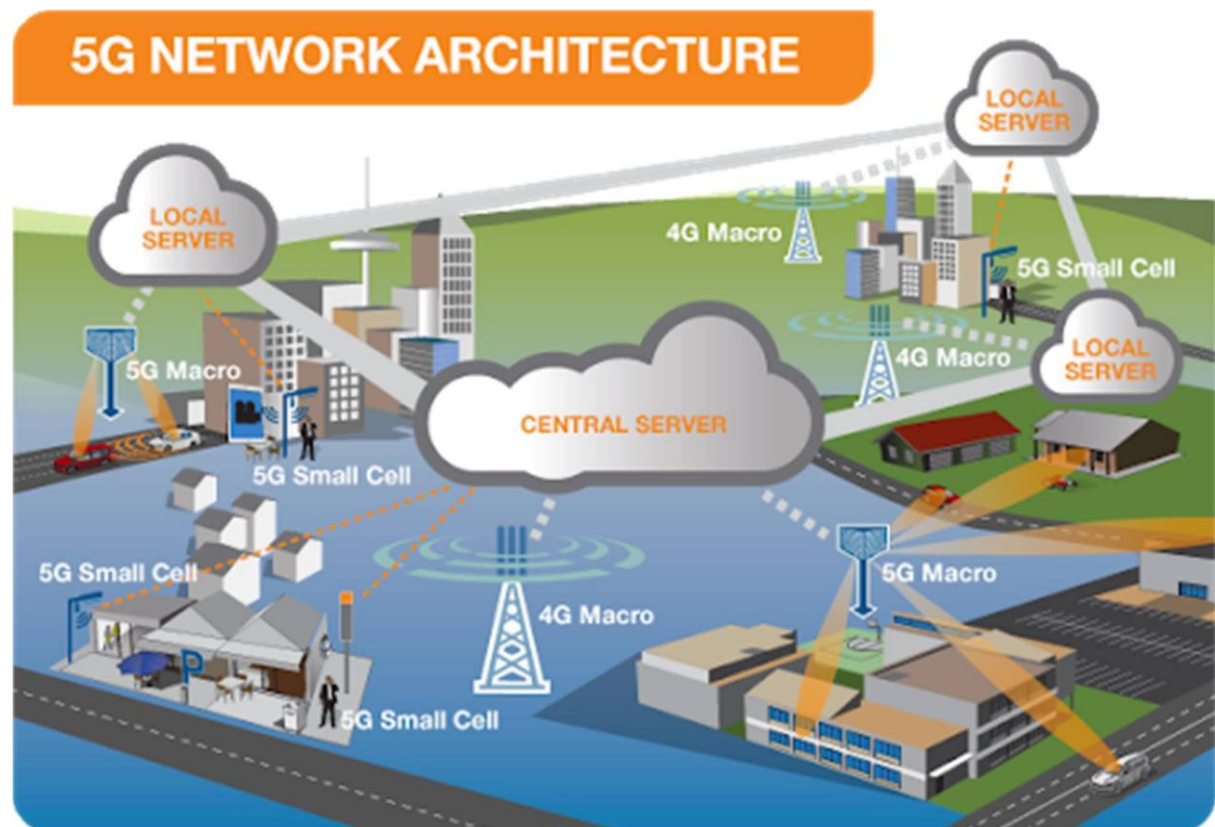


Figure 4.7 5G Network Architecture

1.3.4 Transforming industries

5G is the foundation for flexible, efficient and responsible business. Production lines autonomously reacting to supply and demand Digital replicas that can warn about real machinery faults ahead of time Logistic networks autonomously routing goods based on real-world conditions Full traceability down to the individual item at warehouses and ports Remote access to powerful robots and vehicles for improved safety in risky environments. Increased use of IoT in agriculture to efficiently grow crops.

1.3.5 Elevating experiences

5G sets the stage for more immersive entertainment and more engaging education. Greater realism in VR, AR and extended reality (XR) with lighter devices. Delivering sensory experiences, like touch, through devices. More engaging methods of teaching through immersive content. Immersive virtual meetings to boost remote team productivity. Stable and reliable connectivity in crowded spaces. New angles and interactions for live and remote event spectators.

1.4 Communication Protocols

Communication protocols assist varied network devices to converse with each other by transmitting the analog signals, digital signals, different files & process the data from one device to other devices.

These types of protocols are applicable in telecommunication & computer networks where suitable rules are executed to transmit information from source to destination.

The most vital protocols within networking are TCP (Transmission Control Protocol) & User datagram protocol (UDP).

TYPES OF ELECTRONIC COMMUNICATION PROTOCOLS

Inter System Protocol

Used to communicate between two different devices

Intra System Protocol

Used to communicate between two devices within the circuit board

UART PROTOCOL

UART stands for A Universal Asynchronous Transmitter And Receiver. UART Protocols is a serial communication with two wired protocols. The data cable signal lines are labeled as Rx and Tx. Serial communication is commonly used for transmitting and receiving the signal. It transfers and receives the data serially bit by bit without clock pulses. The UART takes bytes of data and sends the

individual bits in a sequential manner. UART is a half-duplex protocol. Half-duplex means transferring and receiving the data but not at the same time. Most of the controllers have hardware UART on board. It uses a single data line for transmitting and receiving the data. It has one start bit, 8-bit data and a one-stop bit mean the 8-bit data transfer one's signal is high to low. Ex: Emails, SMS, Walkie-talkie. USART stands for A Universal Synchronous and Asynchronous Transmitter and Receiver. It is a serial communication of a two-wire protocol. The data cable signal lines are labeled as Rx and TX. This protocol is used to transmitting and receiving the data byte by byte along with the clock pulses. It is a full-duplex protocol that means transmitting and receiving data simultaneously to different board rates. Different devices communicate with microcontroller to this protocol.

Ex:-Telecommunications.



Figure 4.8 Types of USB

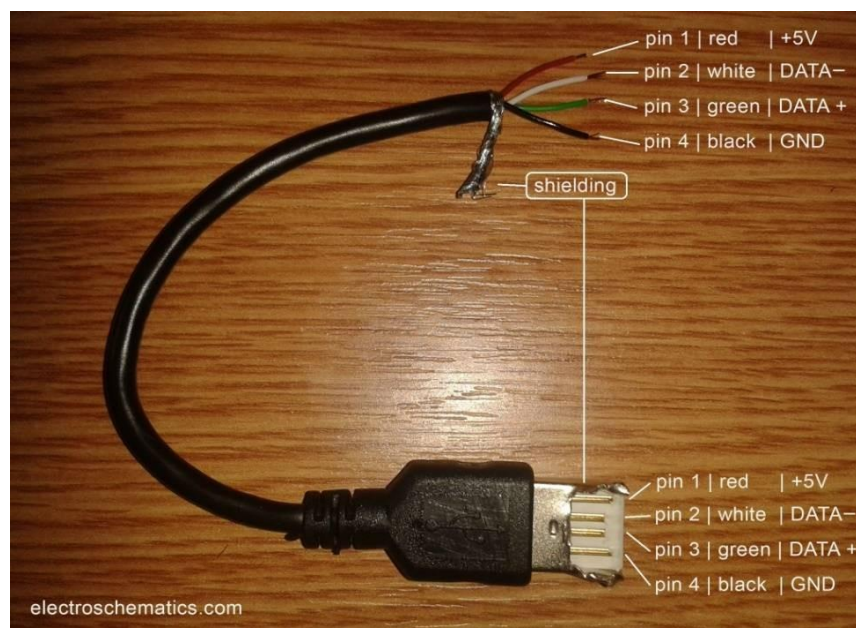


Figure 4.9 USB Cable wiring

DIFFERENCES BETWEEN THE INTER SYSTEM PROTOCOLS

UART	USART	USB
1. UART stands for Universal Asynchronous transmitter and receiver.	1. USART stands for universal synchronous and asynchronous Data transmitter and receiver	1. USB stands for universal serial bus.
2. It is a two wire protocol Rx and TX.	2. It is a two wire protocol Rx and TX.	2. It is a two wire protocol D+ and D-.
3. It is transmitting and receiving packets of The data byte by byte without classes pulses	3. It is send and receives a block of data along with classes pulses.	3. Send and receives a data along with clock pulses
4. It is a half duplex communication	4. It is a full duplex Communication	4. It is also full-communication
5. It is slow compare with : USART	5. It is slow compare with USB	5. It is fast compare with USART and USB

DIFFERENCES BETWEEN THE INTRA SYSTEM PROTOCOLS

I2C	SPI	CAN
1. I2C stands for inter Integrated circuit.	1. SPI stands for serial peripheral Interface.	1. CAN stands for controller Area network.
2. It is developed by the Philips.	2. It is developed by the Motorola.	2. It is developed by the Robert Bosch.
3. It is a Half Duplex protocol	3. It is a full duplex protocol	3. It is a full duplex protocol
4. Synchronization	4. Synchronization	4. Synchronization
5. It is two wire protocols SCL and SDL.	5. It is a four wire protocol SCL and MISO/MOSI, SS	5. It is a two wire protocol CAN H+ and CAN H-.
6. It is multi master protocol	6. It is single master protocol	6. It is multi master protocol
7. With in the circuit board	7. With in the circuit board	7. With in two circuit board.

1.5 Integration of Sensors in Robots and Artificial Intelligence

A combination of various sensors allows an AI robot to determine size, identify an object and determine its distance. Radio-frequency identification (RFID) are wireless sensor devices that provide identification codes and other information. Force sensors provide the ability to pick up objects.

AI and computer vision technologies can help robots to identify and recognize objects they encounter, help pick out details in objects and help with navigation and avoidance. AI-enabled manipulation and grasping. Long considered a difficult task for robots, AI is being used to help robots with grasping items.

Just like sense organs are important in the human body, electronic sensors play a vital role in artificial intelligence. AI work programming intelligence or making computers behave like humans. AI algorithms are used to control robots

AI in robotics helps robots perform the crucial tasks with a human-like vision to detect or recognize the various objects. A huge amount of datasets is used to train the computer vision model, so that robotics can recognize the various objects and carry out the actions accordingly with right results.

Sensors used in AI robots are the same as, or are similar to, those used in other robots. Fully-functional human robots with AI algorithms require numerous sensors to simulate a variety of human and beyond-human capabilities. Sensors provide the ability to see, hear, touch and move like humans.

Sensors in robot allow it to react with its environment in a flexible way. With the help of sensors, robots are able to see and feel, and this would help the robot to perform complex tasks. In order to control their own actions, robots need to know information about the position and the movement of its body and parts.

Robots also have senses in the form of sensors. Sensors are used to evaluate the environment in which the robot is operating and allows the robot to adjust actions based on collected data. Contact sensors—to avoid obstacles. Proximity/Distance sensors—to detect distance of objects in relation to the robot.

Robotic sensors are used to estimate a robot's condition and environment. These signals are passed to a controller to enable appropriate behavior. Sensors in robots are based on the functions of human sensory organs. Robots require extensive information about their environment in order to function effectively.

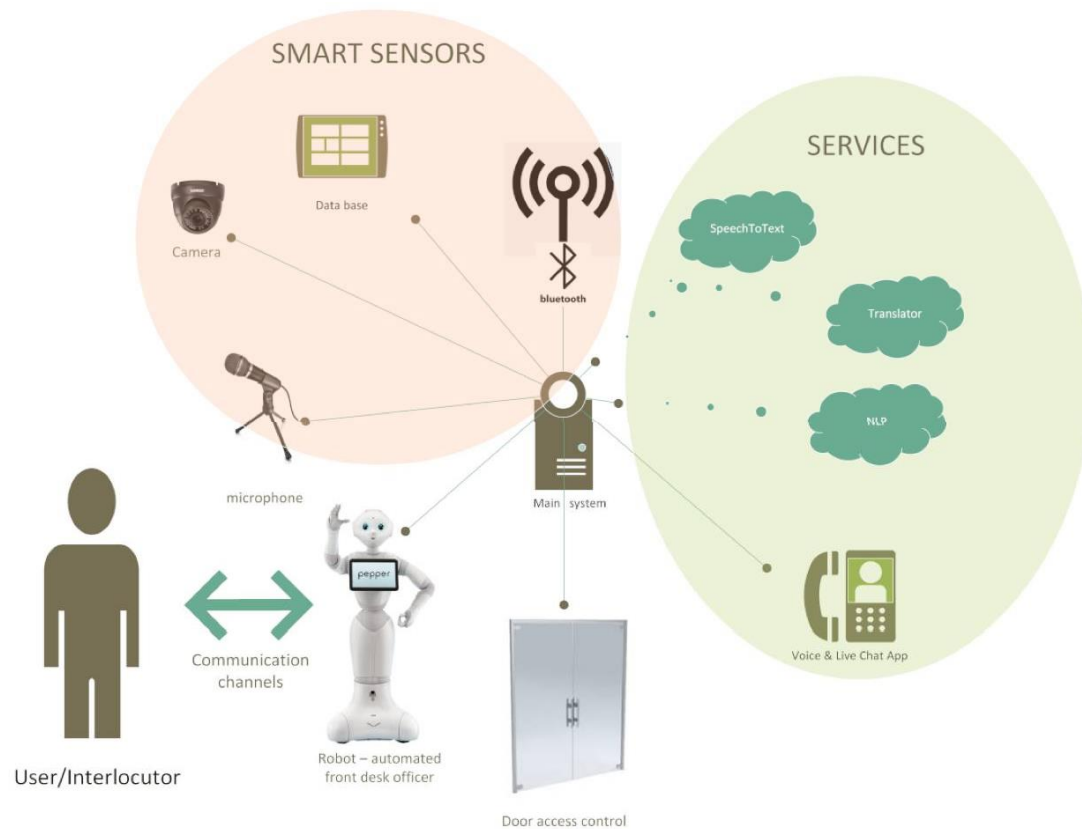


Figure 4.10 Integration of sensors in Robotics

Role of Sensors in AI (Artificial Intelligence)

Just like sense organs are important in the human body, electronic sensors play a vital role in artificial intelligence. AI work programming intelligence or making computers behave like humans. AI algorithms are used to control robots. For example, AI algorithms are used in virtual assistants like Google Assistant, Amazon Alexa and Apple Siri. Without sensors, AI robots are blind, deaf and dumb.

Local Intelligence:

Sensor solutions are mostly responsible for only data acquisition. The raw-data needs to be extracted from the sensor and transmitted to another, more computationally capable device within the network. The receiving end collects the raw-data and performs pre-processing in order to present relevant results. Frequently the raw-data of the IoT device needs to be processed using artificial intelligence, for example as in speech recognition. The demand for artificial intelligence is expected to increase dramatically over the next years since sensor solutions become more complex.

In future, the tendency of processing data within the cloud will be additionally shifted back to local on-device processing. That allows more complex sensor solutions, which include sensor fusion or pattern recognition.

For that kind of applications, local intelligence of the IoT device is needed. Sensor solutions will become resourceful, as they already deliver finalized meaningful data. But, computing elaborate AI solutions within an IoT device, need new solutions which meet power, speed and size constraints. In order to archive this, the trend is shifting to integrated circuits optimized for machine learning. This type of processing is commonly mentioned as edge AI.

1.6 Plant Safety and Security (Including AR and VR safety applications)

Augmented reality (AR) and virtual reality (VR) are closely related but not the same. Augmented reality enhances or 'augments' the real world by adding digital elements – visual, auditory, or sensory – to a real-world view.

AR/VR in Cyber Security

An SOC is a focused facility where security specialists monitor, assess and defend against computer security issues. Introducing Virtual Reality (VR) and Augmented Reality (AR) technology into this environment can enhance team performance, which translates into opportunity gains.

Augmented reality can help provide remote assistance to customers as they repair or complete maintenance procedures on products. It's also a valuable training tool to help inexperienced maintenance crew complete tasks and find the correct service and parts information when they are on-site. Augmented Reality (AR) adds digital elements to the camera of your smartphone, creating the illusion that holographic content is a part of a physical world around. Unlike virtual reality (VR), it is not immersed in an artificial environment.

1.7 Facility Management

Facility management (FM) is a profession that encompasses multiple disciplines to ensure functionality, comfort, safety and efficiency of the built environment by integrating people, place, process and technology. In general, facilities managers are responsible for the maintenance and upkeep of an organization's buildings, ensuring that they meet legal

requirements and health and safety standards. Facility managers (FMs) operate across different business functions, working on both a strategic and operational level.

There are two major types of facilities management, Hard FM and Soft FM. Hard FM refers to services relating to the actual structures and systems that make a facility work, and can include fire safety, plumbing, structural, and elevator maintenance.

Facility management (FM) is a profession that encompasses multiple disciplines to ensure functionality, comfort, safety and efficiency of the built environment by integrating people, place, process and technology.



Figure 4.11 Facility Management

1.8 Intellectual Property Rights

Intellectual Property Rights (IPRs) are legal rights that protect creations and/or inventions resulting from intellectual activity in the industrial, scientific, literary or artistic fields. The most common IPRs include patents, copyrights, marks and trade secrets. Intellectual property rights are the rights given to persons over the creations of their minds. They usually give the creator an exclusive right over the use of his/her creation for a certain period of time.

Intellectual property rights are customarily divided into two main areas:

(i) Copyright and rights related to copyright

The rights of authors of literary and artistic works (such as books and other writings, musical compositions, paintings, sculpture, computer programs and films) are protected by copyright, for a minimum period of 50 years after the death of the author.

Also protected through copyright and related (sometimes referred to as “neighbouring”) rights are the rights of performers (e.g. actors, singers and musicians), producers of phonograms (sound recordings) and broadcasting organizations. The main social purpose of protection of copyright and related rights is to encourage and reward creative work.

(ii) Industrial property

Industrial property can usefully be divided into two main areas:

One area can be characterized as the protection of distinctive signs, in particular trademarks (which distinguish the goods or services of one undertaking from those of other undertakings) and geographical indications (which identify a good as originating in a place where a given characteristic of the good is essentially attributable to its geographical origin).

The protection of such distinctive signs aims to stimulate and ensure fair competition and to protect consumers, by enabling them to make informed choices between various goods and services. The protection may last indefinitely, provided the sign in question continues to be distinctive.

Other types of industrial property are protected primarily to stimulate innovation, design and the creation of technology. In this category fall inventions (protected by patents), industrial designs and trade secrets.

The social purpose is to provide protection for the results of investment in the development of new technology, thus giving the incentive and means to finance research and development activities.

A functioning intellectual property regime should also facilitate the transfer of technology in the form of foreign direct investment, joint ventures and licensing.

The protection is usually given for a finite term (typically 20 years in the case of patents). While the basic social objectives of intellectual property protection are as outlined above, it should also be noted that the exclusive rights given are generally subject to a number of limitations and exceptions, aimed at fine-tuning the balance that has to be found between the legitimate interests of right holders and of users.

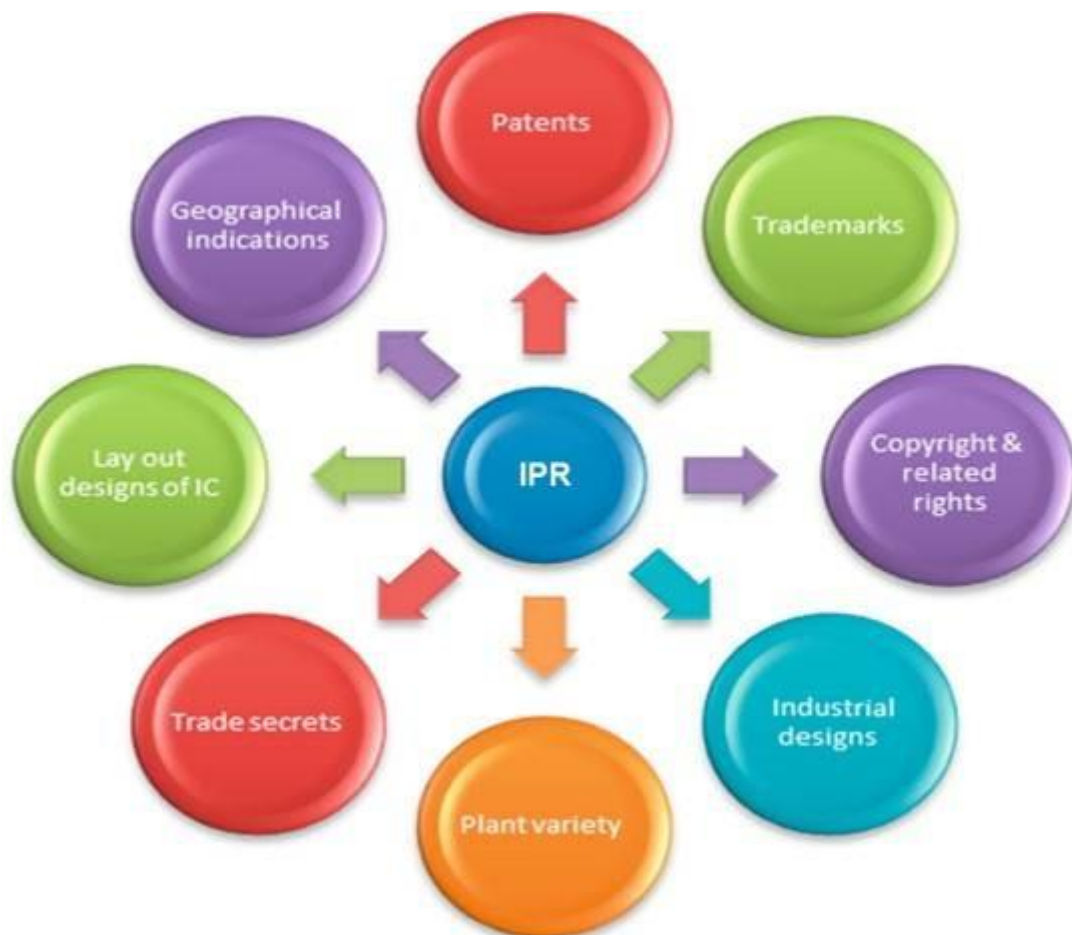


Figure 4.12 IPR

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Question Bank

Part A: Two Marks:

1. What is a sensor question?
2. Why do we need transducers in sensing and actuation?
3. How do sensors and actuators work together?
4. What are the types of sensors?
5. What is meant by inventory management and quality control?
6. Brief about facility management.

Part B:

1. Explain in detail about 5G Technology.
2. Interpret the Integration of Sensors in Robots and Artificial Intelligence.
3. Discuss the Integration of Sensors in Human-Machine Interaction, Industrial IoT.
4. Elaborate the applications of Industry 5.0 in healthcare and power plants.
5. Explain in detail about plant safety and security measures with respect to AR & VR applications.
6. Discuss in detail about Intellectual property rights.